#1 · Professionalism

Who regulates dental hygiene in Canada, and what does the FDHRC actually do?

Answer

Dental hygiene is regulated provincially by a college or regulatory authority in each province, so there is no national regulator and no national licence. The FDHRC develops and administers the NDHCE. It does not register anyone, set scope of practice or discipline registrants.

#2 · Professionalism

Passing the NDHCE makes you what, and what does it not make you?

Answer

It makes you certified, not registered. Certification is one requirement that participating provincial regulators use when deciding on registration. You still apply to the regulator where you intend to work and meet its other requirements before you may practise. Quebec sits outside the NDHCE.

#3 · Professionalism

Under self regulation, whom does a regulatory college owe its duty to?

Answer

The public. A provincial legislature delegated authority to the college on condition that it regulates in the public interest, not in the interest of its registrants. When an item pits the reputation or income of dental hygienists against public safety, public safety wins.

#4 · Professionalism

How does a professional association differ from a regulatory college?

Answer

A college holds statutory authority: it registers practitioners, sets standards, runs quality assurance and handles complaints and discipline. An association such as the CDHA advocates for members. A body named as an association can still hold regulatory authority, as in Saskatchewan, so read the mandate rather than the name.

#5 · Professionalism

Does every province have a standalone dental hygiene college?

Answer

No. Some provinces regulate dental hygienists through a combined oral health regulator or a shared dental board that also covers dentists, denturists and dental technicians. British Columbia amalgamated its dental hygiene college into one combined oral health regulator. Do not assume a standalone college everywhere.

#6 · Professionalism

What defines your scope of practice, and how do you answer when no province is named?

Answer

Provincial legislation and your regulator's standards define what you are legally authorized to perform. Core practice everywhere includes assessment, dental hygiene diagnosis, education, scaling, root planing, polishing and preventive agents. When no province is named, apply the universal rule: work within your legislated scope, your competence and your standards, and refer otherwise.

#7 · Professionalism

Your employer's policy permits a procedure your provincial legislation excludes. What do you do?

Answer

Decline to perform it, state your concern and document it. Employer policy may narrow your legislated scope but never widen it. Following an instruction to work outside your legislated scope is your own professional misconduct, not only the employer's.

#8 · Professionalism

What is self initiation, and is it available everywhere?

Answer

Self initiation is starting a procedure within your scope without an order from a dentist. Ontario permits registrants who meet the College's requirements to initiate scaling below the gum line and root planing, subject to conditions. British Columbia and Alberta also allow it under defined conditions; some jurisdictions still require an order.

#9 · Professionalism

What is a quality assurance program for, and can it be used against you?

Answer

Its purpose is remedial rather than punitive. It usually combines self assessment against the standards, a written learning plan tied to the gaps found, documented continuing education and a periodic practice assessment by random audit. Quality assurance information is kept separate from the complaints stream, so honest self assessment does not expose you to discipline.

#10 · Professionalism

Name the six core ethical principles and say which one outranks the others.

Answer

Autonomy, beneficence, non maleficence, justice, veracity and fidelity. None of them outranks the others; the principles are not ranked, which is why the items are hard. Conflicts are resolved by reasoning through the facts, the people affected and the legal requirements, not by a fixed hierarchy.

#11 · Professionalism

A capable client with untreated periodontitis refuses therapy. Which principle prevails?

Answer

Autonomy prevails over beneficence once a capable client has been told the consequences in plain language. Document the recommendation, the information given and the refusal, then continue to offer care at later visits. A capable client may make a decision you consider unwise.

#12 · Professionalism

You find that a colleague's error harmed a client. Loyalty or truth telling?

Answer

Truth telling. Veracity means the client's right to know prevails over loyalty to a colleague, and it also requires telling a client about your own error. Address the harm, disclose what happened, document the disclosure, and follow the regulator's requirements.

#13 · Professionalism

An ethics item asks what you should do FIRST. What is usually correct?

Answer

Gather information or speak with the client. The structured approach is to gather facts, identify who is affected, name the principles in tension and the legal requirements, generate more than two options, test them against the standards and code of ethics, act, document, then evaluate.

#14 · Professionalism

Is consent a signature or an ongoing process?

Answer

A process. It continues throughout treatment and the client may withdraw it at any moment, including mid procedure. A signed form is evidence that a conversation happened, not the consent itself. Treating without consent can amount to battery, professional misconduct, or both.

#15 · Professionalism

What are the four requirements of valid consent?

Answer

It must relate to the treatment actually proposed, it must be informed, it must be voluntary and free of pressure from the practitioner or the family, and it must come from a person with capacity for that particular decision. Missing any one makes the consent invalid.

#16 · Professionalism

What must a client be told for consent to count as informed?

Answer

The nature of the treatment, expected benefits, material risks and side effects, reasonable alternatives including no treatment, and the cost, with questions answered in language they understand. A material risk is one a reasonable person in that client's position would want to know, so a rare but severe risk still needs disclosure.

#17 · Professionalism

When is implied consent enough, and when must consent be express?

Answer

Implied consent, inferred from conduct such as opening the mouth and reclining, covers only low risk, routine, non invasive steps. Anything invasive, anything carrying a material risk, anything involving radiation and anything with a significant fee needs express consent, given verbally or in writing.

#18 · Professionalism

Your client speaks little English and her adult son offers to interpret. What do you do?

Answer

Use a qualified interpreter instead, and record in the chart who interpreted. A relative may filter the information, may not know clinical terms, and may pressure the client, which undermines both the informed and the voluntary requirements of valid consent.

#19 · Professionalism

How is capacity judged, and does a dementia diagnosis remove it?

Answer

Capacity is decision specific and can fluctuate. It means understanding the information relevant to that decision and appreciating the foreseeable consequences of deciding either way. Capacity is presumed in adults, and dementia or an intellectual disability does not by itself remove it. Someone may consent to fluoride yet lack capacity for a complex plan.

#20 · Professionalism

A 15 year old attends alone and wants sealants placed. Who consents?

Answer

Most Canadian provinces outside Quebec use the mature minor approach rather than a fixed age, so a young person who understands the treatment and its consequences may consent. Ontario's consent legislation sets no minimum age. Some jurisdictions set statutory ages for particular purposes, so check the local rule; otherwise a parent or guardian consents.

#21 · Professionalism

An adult client lacks capacity. Who authorizes care, and on what basis?

Answer

A substitute decision maker taken from the province's ranked list, which usually begins with a court appointed guardian or an attorney under a power of attorney for personal care, then spouse, children, parents and other relatives. The substitute follows the client's previously expressed capable wishes, and otherwise acts in their best interests.

#22 · Professionalism

Can a substitute decision maker authorize care the client refused while capable?

Answer

No. A substitute cannot authorize treatment that the client had already refused while capable. Previously expressed capable wishes govern whenever they are known, and the best interests test applies only where no such wish covers the decision.

#23 · Professionalism

Which privacy law governs client records in a private dental practice?

Answer

The federal PIPEDA applies to personal information handled in the course of commercial activity, which reaches many private practices. Where a province has health privacy legislation declared substantially similar, that provincial statute governs the health sector instead, as Ontario's Personal Health Information Protection Act does.

#24 · Professionalism

What core duties does health privacy legislation place on a practice?

Answer

Collect only what you need, use it only for the purpose it was collected for, secure it, give clients access to their own record, and report significant breaches to the commissioner and to the affected clients. Most real breaches are mundane, such as discussing a client within earshot of others.

#25 · Professionalism

When may client information be disclosed without consent?

Answer

Where a statute compels reporting, where a court orders production, where there is a serious and imminent risk of harm to the client or an identifiable other person, for communicable disease reporting, and to a regulator conducting an investigation. Tell clients about these limits early and disclose only the minimum necessary.

#26 · Professionalism

You suspect a child client is being abused. What exactly must you do?

Answer

Report directly to the child protection authority or the police yourself. The threshold is reasonable suspicion, not proof, and the duty belongs to whoever forms it. You do not investigate first, you do not need parental consent, and telling the dentist does not satisfy the duty.

#27 · Professionalism

Which chairside findings should raise suspicion of child abuse or neglect?

Answer

Head and neck injuries, unusual bruising patterns, a torn labial frenum in a non ambulatory infant, and untreated rampant decay alongside repeated broken appointments. Good faith reporting is protected by law, and professionals face penalties for failing to report.

#28 · Professionalism

A client tells you another regulated health professional touched her sexually. What now?

Answer

Report to the registrar of that professional's college. In Ontario the duty arises when, in the course of practising, you obtain reasonable grounds to believe a regulated health professional has sexually abused a client, and you do not name the client without their consent. The definition covers sexual touching and remarks of a sexual nature.

#29 · Professionalism

A colleague arrives smelling of alcohol before seeing clients. What is the right response?

Answer

Act rather than ignore it. Speak with the colleague when the risk is low and remediable, and report to the regulator when clients are at risk or the behaviour continues. Doing nothing is never correct. Employers generally must also report a registrant terminated, suspended or resigning for conduct, competence or capacity reasons.

#30 · Professionalism

Why can no client consent make a sexual relationship with a current client acceptable?

Answer

The therapeutic relationship is unequal because you hold clinical knowledge, private information and physical access to the client's body. That power imbalance makes genuine consent impossible, so any sexual relationship with a current client is prohibited outright. Crossings often start small, such as a valuable gift or a social media connection.

#31 · Professionalism

You practise in a small town and your neighbour books an appointment. Is that permitted?

Answer

Yes. Dual relationships are not forbidden, and in rural communities they are unavoidable. Recognize the second role, keep clinical objectivity, and refer the person elsewhere when the second role would compromise care or the client's freedom to refuse treatment.

#32 · Professionalism

When does a conflict of interest exist, and what do you do about it?

Answer

It exists whenever a personal interest could reasonably be seen to influence a professional judgment, whether or not you acted on it. Commissions on retail products, referral fees and employer treatment targets all qualify. Avoid the conflict where possible, disclose it where you cannot, and never let it determine the recommendation.

#33 · Professionalism

A dentist orders a procedure you believe is contraindicated for this client. What do you do?

Answer

Raise the concern, document it, and decline to proceed if proceeding would harm the client. Professional accountability means you answer personally for your own clinical decisions and cannot transfer them to a supervisor or employer. Being told to do it is not a defence.

#34 · Professionalism

You spot a wrong entry you made in a chart last week. How do you fix it?

Answer

Never alter or erase the original. Strike through the error so it stays readable, add the correction, then date and initial it. Entries are made at the time of care, and a late entry is identified as a late entry. Falsifying a record is professional misconduct.

#35 · Professionalism

Beyond treatment performed, what belongs in the client record?

Answer

What you assessed, found and recommended, the materials and doses used, what the client was told, and what they agreed to or refused. Record refusals and missed appointments as carefully as treatment. An absent entry is generally treated as care that did not happen.

#36 · Professionalism

How long are dental hygiene client records normally kept?

Answer

Retention periods are set provincially. They commonly run about ten years from the last entry for an adult, and for a child about ten years from the date the child reaches the age of majority. Check the requirement in the province where you practise.

#37 · Professionalism

Walk through what happens after someone complains about you to your regulator.

Answer

The written complaint goes to the registrar, you are notified and may respond, an investigator may copy records, and a screening committee reviews the file. It may take no action, give advice or a caution, require remediation, or refer the matter to a discipline hearing. Ignoring letters from your college makes it worse.

#38 · Professionalism

How does a negligence action differ from a regulatory complaint?

Answer

Negligence is a civil matter decided by a court. The client must show a duty of care, care falling below the standard of a reasonably prudent dental hygienist in similar circumstances, causation of the injury, and real damage. Regulatory and court findings are independent, and one incident can produce both.

#39 · Professionalism

A client with hepatitis C asks for scaling and your employer suggests refusing. What applies?

Answer

Refusing care because of a bloodborne infection is discrimination under the human rights code and is clinically pointless, since routine practices already treat every client as potentially infectious. The duty to accommodate requires adjusting practice up to undue hardship, judged on cost, outside funding and safety, not on inconvenience.

#40 · Professionalism

How does cultural safety differ from cultural awareness, and what do the TRC calls to action ask of you?

Answer

Awareness notices difference and sensitivity respects it. Cultural safety goes further: you examine your own power, bias and assumptions, and the client decides whether the encounter felt safe. TRC calls to action ask that Indigenous healing practices be respected, that professionals be trained in cultural competency, and that students study residential schools and anti racism.

#41 · Evidence Informed Practice

What three elements must an evidence informed decision combine?

Answer

Best available research evidence, the clinician's own expertise from training and experience, and the client's values, circumstances and preferences. Ignoring any one is not evidence informed. Canadian documents prefer evidence informed over evidence based because research is one input into professional judgment, not a cookbook protocol applied to every client.

#42 · Evidence Informed Practice

What does each letter of PICO stand for?

Answer

P is the population or problem described precisely, I is the intervention, exposure or test, C is the comparison such as usual care, a placebo or doing nothing, and O is the outcome stated in measurable terms. Some questions have no meaningful comparison, so C may be left out.

#43 · Evidence Informed Practice

Turn a vague dry mouth worry into a PICO question.

Answer

In adults over 65 taking multiple xerostomic medications, does a prescription 5000 ppm fluoride toothpaste, compared with a standard 1450 ppm toothpaste, reduce new root caries lesions over 12 months? The outcome is countable. Better oral health is not a usable outcome.

#44 · Evidence Informed Practice

Rank the hierarchy of evidence from strongest to weakest.

Answer

Systematic reviews and meta-analyses, then randomized controlled trials, cohort studies, case control studies, cross sectional studies, case reports and case series, with expert opinion at the bottom. The ranking is only a default. A well conducted cohort study beats a tiny randomized trial with heavy dropout.

#45 · Evidence Informed Practice

How does a meta-analysis differ from a systematic review?

Answer

A systematic review states a focused question, searches by a pre-declared protocol, appraises every included study and synthesizes the findings. A meta-analysis goes further and pools the results into one summary estimate with a confidence interval. Pooling weak trials gives a precise looking answer built on weak material.

#46 · Evidence Informed Practice

Why can a randomized controlled trial support a causal claim?

Answer

Allocation by chance distributes known and unknown confounders evenly between the groups, and allocation happens before the outcome occurs, so exposure clearly precedes effect. Alternating clients between groups or assigning by day of the week is not randomization, because the next allocation is predictable.

#47 · Evidence Informed Practice

What can a cohort study establish, and what limits it?

Answer

A cohort follows a defined group forward in time, comparing exposed with unexposed people. It measures incidence and shows that exposure preceded the outcome, so it suits harm and prognosis questions where randomizing people to harm would be unethical. Assignment was not random, so residual confounding remains possible.

#48 · Evidence Informed Practice

When is a case control design the right choice?

Answer

For rare diseases. It starts with the outcome and looks backwards, comparing past exposures in people who have the disease against a similar group who do not. It reports an odds ratio rather than a relative risk because it cannot measure risk, and recall bias is a constant threat.

#49 · Evidence Informed Practice

What can a cross sectional study give you?

Answer

It measures exposure and outcome at the same moment, so it gives prevalence and shows associations. It cannot show which came first, so causation is out of reach. Cross sectional surveys are useful for sizing a community's disease burden and planning services.

#50 · Evidence Informed Practice

Distinguish confounding, selection bias and recall bias.

Answer

Confounding is a third variable linked to both exposure and outcome, as smoking is to periodontal disease. Selection bias means the people studied differ systematically from the population the result is applied to, such as motivated clinic volunteers. Recall bias means people with the disease remember past exposures differently.

#51 · Evidence Informed Practice

Why does intention to treat analysis matter after dropout?

Answer

Dropout is rarely random. If clients left the treatment group because the product tasted unpleasant, analyzing only those who remained flatters the treatment. Intention to treat counts every participant in the group they were originally allocated to, whether or not they finished. Per protocol analysis includes only compliers and exaggerates benefit.

#52 · Evidence Informed Practice

What do allocation concealment and blinding each protect?

Answer

Allocation concealment stops the enroller from seeing which group comes next, protecting randomization itself. Blinding protects the measurement: in a single blind study the participant does not know the assignment, in a double blind study the outcome assessor does not either. When the operator cannot be blinded, blind the examiner.

#53 · Evidence Informed Practice

State the sensitivity formula and work through one example.

Answer

Sensitivity equals TP divided by (TP plus FN), the proportion of people who truly have the disease that the test correctly identifies. If 200 adults truly have active disease and the test is positive in 170 of them, sensitivity is 170 divided by 200, which equals 0.85, or 85 percent.

#54 · Evidence Informed Practice

State the specificity formula and work through one example.

Answer

Specificity equals TN divided by (TN plus FP), the proportion of people free of the disease that the test correctly clears. If 800 adults are truly healthy and 720 test negative, leaving 80 false positives, specificity is 720 divided by 800, which equals 0.90, or 90 percent.

#55 · Evidence Informed Practice

How are positive and negative predictive value calculated?

Answer

They read across the rows of the two by two table. With 170 true positives and 80 false positives, positive predictive value is 170 divided by 250, which equals 68 percent. With 720 true negatives and 30 false negatives, negative predictive value is 720 divided by 750, which equals 96 percent.

#56 · Evidence Informed Practice

Why does one test perform worse in a low risk population?

Answer

Predictive values depend on prevalence, while sensitivity and specificity are properties of the test. Applying an 85 percent sensitive, 90 percent specific test to 2000 adults with 5 percent disease yields 85 true positives and 190 false positives, so positive predictive value falls to 85 divided by 275, about 31 percent.

#57 · Evidence Informed Practice

Contrast prevalence and incidence using a worked example.

Answer

Prevalence is existing cases at a point in time; incidence is new cases arising in a previously unaffected group over a defined period. Of 5000 adults, 1500 have periodontitis, so prevalence is 30 percent. Following the remaining 3500 for one year, 175 new cases give an incidence of 5 percent.

#58 · Evidence Informed Practice

Calculate relative risk and relative risk reduction from trial data.

Answer

Control group risk is 60 of 200, which equals 0.30, and varnish group risk is 40 of 200, which equals 0.20. Relative risk is 0.20 divided by 0.30, which equals 0.67, and a value below 1 indicates benefit. Relative risk reduction is 1 minus 0.67, which equals 33 percent.

#59 · Evidence Informed Practice

Calculate absolute risk reduction and number needed to treat.

Answer

Absolute risk reduction is the plain difference in risk: 0.30 minus 0.20 equals 0.10, or 10 percentage points. Number needed to treat equals 1 divided by the absolute risk reduction, so 1 divided by 0.10 equals 10. Ten children must receive the programme for one extra child to avoid a lesion.

#60 · Evidence Informed Practice

What does a p value tell you, and what does a confidence interval add?

Answer

A p value is the probability of a result at least as extreme if there were truly no difference, so p of 0.03 means 3 times in 100 by chance. A confidence interval adds the plausible range and the precision; a relative risk interval of 0.52 to 0.86 excludes 1, so it is significant.

#61 · Evidence Informed Practice

Give an example of a statistically significant but clinically trivial result.

Answer

A trial of 5000 participants shows a mouthrinse reducing mean probing depth by 0.05 mm with p equal to 0.001. The finding is real and trivial, because 0.05 mm sits below the measurement error of a periodontal probe. Very large samples make tiny differences reach statistical significance.

#62 · Evidence Informed Practice

Why does a correlation between two variables not prove causation?

Answer

Two variables can move together because one causes the other, because a third factor drives both, or by coincidence. Coffee drinking correlates with periodontal disease largely because smoking travels with it. A causal claim needs controlled allocation, or evidence of consistency, dose response, biological plausibility and correct temporal order.

#63 · Evidence Informed Practice

What is TCPS 2, and what does a Research Ethics Board do?

Answer

TCPS 2 is the Canadian Tri-Council Policy Statement on ethical conduct for research involving humans, issued by CIHR, NSERC and SSHRC. Its principles are respect for persons, concern for welfare and justice. A Research Ethics Board approves research before it begins and monitors it, and the board decides on delegated review.

#64 · Evidence Informed Practice

Name the warning signs of a predatory journal.

Answer

Unsolicited invitations to submit, a title mimicking a well known journal, a scope spanning unrelated fields, an unverifiable editorial board, no listing in a recognized index, a publication fee and acceptance within days with little peer review. Treat such an article as an unverified claim.

#65 · Evidence Informed Practice

How do you proceed when two guidelines on one topic conflict?

Answer

Check the publication dates, since the newer document may include trials the older one could not. Compare the populations studied, and compare study quality rather than counting studies. Look for a recent systematic review, check Canadian professional bodies and the provincial regulator, then share any remaining uncertainty with the client.

#66 · Communication

What is the difference between open and closed questions?

Answer

Open questions invite an answer of any length and produce information you did not know to ask for. Closed questions get a yes, no or single fact, which suits medical history but yields thin data. Open wide early, then narrow down to confirm details.

#67 · Communication

Name the core therapeutic communication skills beyond questioning.

Answer

Active listening means visible attention: stop charting, turn toward the client, do not interrupt. Reflection feeds back what was heard, as content or as feeling. Summarizing gathers several threads and hands them back. Silence gives the client room to think, so do not fill it with reassurance.

#68 · Communication

How do empathy and sympathy differ in client care?

Answer

Sympathy is feeling for the client from outside their situation and produces pity or hurried comfort, which closes the conversation. Empathy is understanding the experience from the client's own frame of reference and reflecting it back without judgment, which keeps the client talking.

#69 · Communication

Client: "They told me I might lose this tooth." Best response?

Answer

Reflect the feeling: "Being told you might lose that tooth must be hard to hear." That is empathy, and it invites the client to say more. "Don't worry, this happens to everyone" is sympathy plus false reassurance, and it shuts the conversation down.

#70 · Communication

What is health literacy, and why is low health literacy easy to miss?

Answer

Health literacy is the ability to find, understand and use health information to make decisions, and it is not the same as general literacy or intelligence. Roughly six in ten Canadian adults sit below the level they need. Shame hides it, so never rely on "Does that make sense?"

#71 · Communication

What does plain language look like in a dental hygiene appointment?

Answer

Use everyday words, or use the clinical word and define it at once: "gingivitis, which is gum inflammation." Keep sentences short and limit the visit to two or three key messages. Handouts sit near a Grade 6 reading level, with headings and captioned pictures.

#72 · Communication

What is teach back, and how is it framed?

Answer

The client explains or demonstrates the instruction in their own words, framed as a check on your teaching rather than a test of the client. If the answer is wrong, re-teach in different words and ask again. Show back, a physical demonstration, is stronger for anything manual.

#73 · Communication

Client nods and says "Yes, that all makes sense." What next?

Answer

Do not accept the nod. Ask for teach back: "I want to be sure I explained it clearly. Can you show me how you will use the interdental brush at home?" Agreement is the polite answer and proves nothing about understanding.

#74 · Communication

What is motivational interviewing, and what is its spirit?

Answer

A collaborative, goal oriented conversation style that strengthens the client's own motivation to change. Its spirit is partnership, acceptance, compassion and evocation, meaning the reasons for change are drawn out of the client. It is not persuasion, confrontation or scare tactics, and not a route to the plan you already made.

#75 · Communication

What does OARS stand for in motivational interviewing?

Answer

Open questions, Affirmations, Reflective listening and Summaries. Affirmations name genuine specific strengths rather than generic praise. Reflective listening is the workhorse skill, so a good session holds more reflections than questions. Summaries pull the client's own reasons for change together.

#76 · Communication

What is change talk, and what does the hygienist do with it?

Answer

Change talk is any client speech arguing for change: desire, ability, reasons, need, commitment and taking steps. Sustain talk argues for the status quo. Listen for change talk, ask for more of it, reflect it and summarize it, while not amplifying sustain talk.

#77 · Communication

Client: "My grandmother never flossed and kept her teeth." Best response?

Answer

Do not counter argue, because that makes the client defend the behaviour out loud and entrench it. Reflect instead: "It seems this may not feel worth the effort." Reframing, emphasizing personal choice and shifting focus also work. This is rolling with resistance.

#78 · Communication

What is the righting reflex, and how does motivational interviewing differ from advice?

Answer

The righting reflex is the professional urge to fix the client by telling them what to do, and motivational interviewing asks you to hold it back. Simple advice is one directional: name the problem, state the solution, expect compliance. Choose motivational interviewing when the client is ambivalent or unchanged despite repeated instruction.

#79 · Communication

Name the stages of change, and say what relapse means.

Answer

Precontemplation, contemplation, preparation, action and maintenance, from the transtheoretical model of Prochaska and DiClemente. Relapse is a return to an earlier stage and is expected rather than exceptional. Normalize it, affirm what worked before, and re-engage instead of treating it as failure.

#80 · Communication

Client: "My gums bleed but I don't see a problem." Stage and action?

Answer

This is precontemplation. Raise awareness without pressure and offer information with permission rather than pushing a home care plan. Setting a specific flossing goal and picking an aid belong to preparation, and would miss where this client actually is.

#81 · Communication

How do you match the intervention to contemplation and to preparation?

Answer

In contemplation the client is ambivalent, so explore both sides and draw out reasons for change in the client's own words; motivational interviewing does most of its work here. In preparation the client intends to act soon, so choose one behaviour, set a realistic goal, pick the aid, and plan when it happens.

#82 · Communication

What supports a client in the action and maintenance stages?

Answer

In action the client is performing the new behaviour, so support, affirm and reinforce at short recall intervals. In maintenance the behaviour has been sustained, so identify high risk situations and consolidate it into routine. Neither stage needs the awareness raising that suits precontemplation.

#83 · Communication

How should the hygienist communicate with a young child and caregiver?

Answer

Speak to the child directly at their level in short, concrete, positive words, and use tell show do: explain, demonstrate on a finger or model, then perform. Use positive reinforcement, never frightening words for instruments. The caregiver receives the technical explanation and gives consent, and caregiver anxiety needs addressing too.

#84 · Communication

What are the rules for communicating with an older adult?

Answer

Do not assume hearing loss or cognitive decline from age alone. Allow more time, and watch for polypharmacy, xerostomia and dexterity limits when planning home care. Speak to the older adult rather than the accompanying family member, even when that person answers first.

#85 · Communication

How do you adapt communication for hearing loss and for vision loss?

Answer

For hearing loss, get attention first, face the client with light on your face, remove the mask at a safe distance, speak at normal pace and volume, and rephrase rather than repeat. For vision loss, announce arrival and departure, describe each step before doing it, and offer a guiding arm.

#86 · Communication

How is communication adapted for a client with cognitive impairment?

Answer

Reduce distraction, give short single step instructions, allow extra time, and rely on repetition and demonstration rather than complex verbal explanation. Involve the caregiver for history and consent under provincial substitute decision maker rules, while still addressing the client directly and seeking assent for each step.

#87 · Communication

What does trauma informed care look like in the operatory?

Answer

It rests on safety, trustworthiness, choice, collaboration and empowerment. Explain before doing, offer a stop signal that is always honoured, avoid restraint, allow a semi upright position, and never show surprise or judgment about the state of the mouth. Do not probe for details of the trauma.

#88 · Communication

When is a trained interpreter required, and why not a family member?

Answer

Use a trained interpreter whenever the shared language is too weak for informed consent. Speak directly to the client in the first person, in short segments, without idioms, and document that an interpreter was used. Family members, above all children, edit and soften what is said and compromise confidentiality.

#89 · Communication

How do cultural humility and cultural safety differ?

Answer

Cultural humility is a lifelong process of self reflection on your own assumptions and on the power imbalance in the professional relationship. Cultural safety is defined by the person receiving care: it is safe only if the client experiences it as free of racism and judgment. The provider cannot declare their own care safe.

#90 · Communication

How are unwelcome findings and clinical errors communicated?

Answer

Deliver findings upright and privately: warn briefly, state the finding plainly, allow silence, ask permission before adding detail, and finish with a clear next step. An error is disclosed promptly in plain words with an apology, the consequences, what is being done, and a factual record. Concealment is worse than the error.

#91 · Collaboration

What is interprofessional collaboration?

Answer

Two or more providers from different professions working with a client, and often the family, toward shared goals through one shared plan. The client is a member of the team rather than a subject of it.

#92 · Collaboration

How does multidisciplinary care differ from interprofessional care?

Answer

Multidisciplinary care means several professions each do their own part in parallel. Interprofessional care means those professions build one plan together, agree who does what, and report results back to each other.

#93 · Collaboration

Name the six CIHC interprofessional competency domains.

Answer

Role clarification, team functioning, client, family and community centred care, collaborative leadership, interprofessional communication, and interprofessional conflict resolution. They come from the Canadian Interprofessional Health Collaborative framework.

#94 · Collaboration

What does role clarification require of a dental hygienist?

Answer

Describing your own scope accurately and describing the scope of other professions accurately. It is a safety issue, not a courtesy, because overlapping roles let every member assume someone else completed a task that nobody did.

#95 · Collaboration

By what concrete mechanisms does collaboration improve client outcomes?

Answer

It reduces duplicated assessments, catches contraindications before treatment rather than after, closes gaps at transitions of care where most errors happen, and shortens the delay between a suspicious finding and a diagnosis. Goodwill alone is not a mechanism.

#96 · Collaboration

List the predictable barriers to interprofessional collaboration.

Answer

Unclear roles, hierarchy that discourages junior members from speaking, incompatible record systems, no protected time for team meetings, and assuming that silence means agreement. These barriers appear often as exam distractors.

#97 · Collaboration

What is the core scope of the dental hygienist role?

Answer

Assessment, dental hygiene diagnosis, planning, periodontal debridement and scaling, preventive and therapeutic agents, oral health education, and evaluation of outcomes. Hygienists are primary oral health care providers regulated provincially.

#98 · Collaboration

How does a dental hygiene diagnosis differ from a dentist's diagnosis?

Answer

A dental hygiene diagnosis names conditions you are educated and authorized to treat, such as generalized moderate plaque induced gingivitis with contributing xerostomia. The dentist diagnoses oral disease and pathology such as caries. The two are distinct, not competing.

#99 · Collaboration

Is dental hygiene scope of practice the same across Canada?

Answer

No. Dental hygiene is regulated provincially, and scope, self initiation and prescribing authority vary between provinces. Several provinces allow independent practice or self initiation subject to conditions. Never answer a scope question with a national rule that does not exist.

#100 · Collaboration

What falls within the dentist's scope of practice?

Answer

Diagnosis and treatment of oral disease across the full range, including caries diagnosis and restoration, endodontics, extractions, prosthodontics and prescribing. Dentists and hygienists overlap on scaling, but diagnosis of pathology belongs to the dentist.

#101 · Collaboration

What is the main structural difference between a dental assistant and a hygienist?

Answer

Assistants support clinical procedures, take radiographs where certified, and manage infection prevention and control. In many provinces certified or level II assistants hold intraoral duties such as rubber dam, fluoride varnish, impressions and polishing, but those duties are delegated and supervised.

#102 · Collaboration

What do dental therapists do, and where do they practise in Canada?

Answer

They provide a defined set of restorative and surgical services, including simple restorations and primary tooth extractions, along with preventive care. Their practice has been concentrated in Saskatchewan and in northern and Indigenous communities. Their training and regulation differ from the hygienist pathway.

#103 · Collaboration

What is a denturist's role, and does a client need a referral?

Answer

Denturists design, construct and fit removable complete and partial dentures. In most provinces they see clients directly without a dentist referral, so an option requiring a referral first is usually wrong.

#104 · Collaboration

What do dental technicians and technologists do?

Answer

They fabricate crowns, bridges and appliances in a laboratory on the written prescription of a dentist or denturist. They have no direct clinical contact with the client, so they never assess or treat.

#105 · Collaboration

Why is overlap between oral health roles a safety concern?

Answer

Overlap is deliberate and efficient. Hygienists, assistants and therapists all apply fluoride varnish, and hygienists and dentists both scale. The risk is diffusion of responsibility: when everyone assumes someone else took the blood pressure, nobody did.

#106 · Collaboration

Can a nurse practitioner answer a dental consultation request?

Answer

Yes. In much of Canada a nurse practitioner is the client's primary provider and can confirm a diagnosis, report control of a chronic condition, supply laboratory values and adjust medications. An option insisting the request go to a physician instead is usually wrong.

#107 · Collaboration

Which professional is the fastest route to accurate medication information?

Answer

The pharmacist. They provide interaction checks and a complete dispensing history, which is the practical answer when a client arrives with a bag of bottles and cannot explain what any of them are for.

#108 · Collaboration

When would you collaborate with a registered dietitian?

Answer

For caries risk driven by dietary sugars, for eating disorders presenting as dental erosion, and for nutrition in clients with impaired chewing. You supply the oral findings and risk assessment; the dietitian builds the dietary plan.

#109 · Collaboration

Why does a speech language pathology assessment change your appointment?

Answer

Speech language pathologists assess dysphagia, tongue and lip function, and clients with cleft palate or acquired brain injury. A client who aspirates is at risk from water in your operatory, so a swallowing assessment changes how you position and suction.

#110 · Collaboration

What access problems do occupational therapists and physiotherapists solve?

Answer

They adapt a toothbrush handle for a client with rheumatoid arthritis, train a caregiver in safe positioning, and advise on wheelchair transfer. These are barriers you cannot solve alone from the operatory.

#111 · Collaboration

What does a social worker contribute to a client's oral health?

Answer

Social workers address the determinants driving most of the disease you see, including income, housing, food security, family violence and transportation. Referring for those determinants treats the cause rather than repeating oral hygiene instruction that the client cannot act on.

#112 · Collaboration

What should you give a personal support worker doing daily mouth care?

Answer

A short written protocol naming the product, the frequency, the technique, and who to call when the mouth bleeds or the client refuses. They deliver the daily care but will not read a periodontal chart.

#113 · Collaboration

Is dental hygiene care safe during pregnancy, and when?

Answer

Yes. Oral care is safe and appropriate in every trimester, and midwives and obstetric providers are your partners. Later in pregnancy use comfort measures such as left lateral positioning. Deferring routine care to after delivery is not the correct answer.

#114 · Collaboration

Why are pediatricians natural collaborators for a hygienist?

Answer

They see infants long before an oral health provider does, which makes them partners for fluoride varnish programs and early childhood caries screening. Collaboration with them moves prevention earlier than the first dental visit.

#115 · Collaboration

Why contact the oncologist before treating a client facing cancer therapy?

Answer

For head and neck radiation, chemotherapy or a stem cell transplant, dental clearance, extraction timing, and management of mucositis and osteoradionecrosis risk are decided jointly. Timing extractions without that discussion can expose the client to serious harm.

#116 · Collaboration

What do you ask a cardiologist and an endocrinologist for?

Answer

The cardiologist confirms cardiac status, anticoagulation and any need for prophylaxis. The endocrinologist confirms diabetes control, including the most recent glycated hemoglobin value and its date. Both answers change what you can safely do at the appointment.

#117 · Collaboration

What is the difference between association and causation?

Answer

Association means two conditions occur together more often than chance predicts. Causation means one condition produces the other. Association can arise from shared risk factors such as smoking, age, obesity and low income, from reverse causation, or from residual confounding.

#118 · Collaboration

Describe the relationship between periodontitis and diabetes.

Answer

It is the strongest oral systemic relationship and it is genuinely bidirectional. Poorly controlled diabetes increases the risk, extent and severity of periodontitis and impairs healing after therapy, while severe periodontitis is associated with worse glycemic control.

#119 · Collaboration

What effect does periodontal therapy have on glycated hemoglobin?

Answer

Intervention trials show non surgical periodontal therapy produces a modest reduction in glycated hemoglobin in people with type 2 diabetes, on the order of a few tenths of a percentage point at three to six months. It supports but never replaces medical management.

#120 · Collaboration

What does the evidence support about periodontitis and cardiovascular disease?

Answer

It is an association, and causation is not established. People with periodontitis have a higher rate of atherosclerotic events and plausible mechanisms exist, but major cardiology and periodontal bodies state the evidence does not show periodontitis causes atherosclerotic cardiovascular disease.

#121 · Collaboration

Does treating periodontitis in pregnancy improve birth outcomes?

Answer

The evidence is mixed and the honest answer is no. Observational studies link periodontitis with preterm birth, low birth weight and pre-eclampsia, but randomized treatment trials have not consistently shown benefit. Those trials did establish that non surgical therapy during pregnancy is safe.

#122 · Collaboration

What should you tell a pregnant client about periodontal treatment?

Answer

Explain that periodontal care during pregnancy is safe and worth having for her own oral health. Do not promise a better birth outcome, because treatment trials have not consistently shown reduced preterm birth or low birth weight.

#123 · Collaboration

Which two further oral systemic links should you know, and how strong are they?

Answer

Aspiration pneumonia in dependent older adults has good intervention evidence, since structured daily oral care in long term care reduces pneumonia incidence and mortality. Rheumatoid arthritis and periodontitis share an association with no established causal direction.

#124 · Collaboration

When should you seek a medical consultation before treating?

Answer

When the health history leaves a question that changes what you will do. Triggers include unclear or uncontrolled diabetes, unstable angina or recent cardiac event, bleeding disorder or anticoagulants with extensive instrumentation, immunosuppression, antiresorptive therapy, uncontrolled hypertension, and any condition that has changed.

#125 · Collaboration

What makes a consultation request useful, and what do you do while waiting?

Answer

Identify yourself and the client with consent, describe the planned treatment in terms a physician understands, give your findings, then ask one closed answerable question rather than writing please advise. While waiting, document the request, deliver only no added risk care, and never assume the reply.

#126 · Collaboration

What does referral transfer, and what stays with the hygienist?

Answer

Referral transfers part of the care to a provider with different expertise. The hygienist remains accountable for the part of the care she keeps, for following up that the client attended, and for continuing her own dental hygiene care within her scope.

#127 · Collaboration

Which findings require urgent rather than routine referral?

Answer

Spreading facial swelling, airway compromise, trismus with fever, uncontrolled bleeding, avulsion of a permanent tooth, signs of stroke or a cardiac event, and any oral lesion suspicious for malignancy. Everything else is referred routinely.

#128 · Collaboration

State the two week rule for a suspicious oral lesion.

Answer

An ulcer, a red, white or mixed patch, an indurated mass or unexplained numbness that persists two weeks after any obvious local cause is removed requires referral for definitive assessment and biopsy. Another recall and another look is the wrong answer.

#129 · Collaboration

What must be documented when a suspicious oral lesion is found?

Answer

Record the site in FDI notation, for example the buccal mucosa adjacent to tooth 36, plus size, colour, borders, texture, symptoms and duration, ideally with a photograph. Also record the referral, the date sent and the date the client attended.

#130 · Collaboration

How should a suspicious lesion referral be explained to the client?

Answer

Explain what was seen, why the referral is being made, and what happens next. Do not tell the client it is probably nothing, because that discourages attendance. Then confirm that the client actually attended the appointment.

#131 · Collaboration

When is referral to a periodontist indicated?

Answer

Rapidly progressing periodontitis, deep sites that do not respond to non surgical therapy, furcation involvement needing surgery, recession requiring grafting, and implant complications. Failure to respond to thorough non surgical therapy is the common exam trigger.

#132 · Collaboration

When is referral to an oral and maxillofacial surgeon indicated?

Answer

Impacted teeth, complex extractions, orofacial trauma, and biopsy or pathology involving bone. A lesion needing surgical biopsy or a hard tissue lesion seen on a radiograph goes to this specialist.

#133 · Collaboration

When is referral to an orthodontist indicated?

Answer

Malocclusion, crowding that contributes to disease because it prevents adequate plaque control, and skeletal discrepancies. The dental hygiene reason for the referral is usually that tooth position is driving gingival inflammation the client cannot control.

#134 · Collaboration

When is referral to an endodontist indicated?

Answer

Suspected pulpal necrosis, persistent apical periodontitis, retreatment of a previously root filled tooth, and difficult canal anatomy. A sinus tract or a tooth that stays symptomatic after debridement points to a pulpal rather than a periodontal source.

#135 · Collaboration

When is referral to a prosthodontist indicated?

Answer

Complex restorative rehabilitation, full mouth reconstruction, and maxillofacial prostheses. Extensive tooth wear, multiple failing restorations or a client needing rebuilt occlusal relationships exceed what routine restorative care manages well.

#136 · Collaboration

Which findings prompt referral to a physician or nurse practitioner?

Answer

Signs of undiagnosed systemic disease the hygienist notices first: repeated elevated blood pressure readings, unexplained weight loss, polydipsia and polyuria suggesting undiagnosed diabetes, and persistent lymphadenopathy. A nurse practitioner can receive and answer these referrals in full.

#137 · Collaboration

Which cardiac conditions still warrant infective endocarditis prophylaxis?

Answer

Only the highest risk group: a prosthetic heart valve or prosthetic material used in valve repair including transcatheter valves, annuloplasty rings and clips; previous infective endocarditis; specific congenital heart disease; and cardiac transplant recipients who develop valvulopathy.

#138 · Collaboration

Which congenital heart disease qualifies for endocarditis prophylaxis?

Answer

Unrepaired cyanotic congenital heart disease; a defect repaired with prosthetic material during the first six months after the procedure; and a repaired defect with residual shunt or valvular regurgitation at or next to prosthetic material.

#139 · Collaboration

Name cardiac conditions that no longer warrant antibiotic prophylaxis.

Answer

Mitral valve prolapse, bicuspid aortic valve, isolated acquired valve disease, a history of rheumatic fever without valve dysfunction, coronary artery stents, coronary bypass surgery, pacemakers and implantable defibrillators. These appear as distractors built on outdated guidance.

#140 · Collaboration

Which dental procedures are covered when prophylaxis is indicated?

Answer

Procedures that manipulate gingival tissue or the periapical region of the tooth, or that perforate oral mucosa. Scaling, periodontal debridement, probing, extractions and intraligamentary injection are all covered.

#141 · Collaboration

Which dental procedures do not require prophylaxis even in a high risk client?

Answer

Routine radiographs, taking impressions, placing orthodontic brackets, adjusting removable appliances, and bleeding from lip trauma. None of these manipulate gingival tissue or the periapical region, or perforate oral mucosa.

#142 · Collaboration

State the standard adult and child antibiotic prophylaxis regimen.

Answer

Amoxicillin 2 g orally for an adult, taken 30 to 60 minutes before the procedure, and 50 mg/kg for a child. The timing matters because the aim is an adequate blood level at the moment of the bacteremia.

#143 · Collaboration

What is used for prophylaxis when the client reports penicillin allergy?

Answer

Doxycycline, azithromycin or clarithromycin, or a cephalosporin such as cephalexin when the allergy history does not include anaphylaxis, angioedema or urticaria. Clindamycin is no longer recommended because of serious adverse effects including Clostridioides difficile infection.

#144 · Collaboration

What if the client cannot swallow the dose, or the dose is missed?

Answer

For a client unable to take oral medication, ampicillin, cefazolin or ceftriaxone is given parenterally. If the dose is forgotten it may still be given up to two hours after the procedure, and this is recorded.

#145 · Collaboration

Do clients with prosthetic joints need routine antibiotic prophylaxis?

Answer

No. Routine prophylaxis before dental treatment for prosthetic joints is no longer generally recommended. Orthopedic and dental organizations concluded the evidence does not support a link between dental procedures and prosthetic joint infection, and that routine antibiotic harms outweigh benefit.

#146 · Collaboration

What is the exception for a client with a prosthetic joint?

Answer

The orthopedic surgeon or physician may still direct prophylaxis for an individual client, for example after a previous joint infection or in significant immunosuppression. Where a written direction exists, follow it and document who gave it rather than overruling it from the chair.

#147 · Collaboration

What does SBAR stand for and when is it used?

Answer

Situation, Background, Assessment and Recommendation. It structures an urgent telephone call or a written request to another provider so the key information arrives in a predictable order and the receiver knows what action is being asked for.

#148 · Collaboration

What is a closed loop readback and what error does it prevent?

Answer

The receiver repeats the instruction back and the sender confirms it is correct. It prevents the classic drug name and dose error, where a misheard medication or number is acted on without anyone noticing the mismatch.

#149 · Collaboration

What is the purpose of a safety huddle?

Answer

A short meeting at the start of the day where the team reviews the schedule and flags risks, such as the client who needs a prophylaxis dose, the client requiring a lift transfer, or a pending consultation reply.

#150 · Collaboration

What five elements belong in every clinical handover?

Answer

Who the client is, what was done, what is outstanding, what the risks are, and what the receiver must do next. Using the same structure every time is what makes handover reliable, because transitions of care are where most errors happen.

#151 · Collaboration

Who attends a long term care conference?

Answer

The resident, family, nursing, the physician or nurse practitioner, dietary, rehabilitation and oral health. The resident and family are members of the team rather than an audience, so the plan is agreed with them.

#152 · Collaboration

What should the hygienist bring to a long term care conference?

Answer

One page giving the oral findings, the daily care protocol, the products, who performs the care, and the trigger for calling the hygienist back. Then get it written into the resident's care plan, because a plan living only in the dental chart is not carried out.

#153 · Collaboration

When must the oncology team be contacted about a client's dental care?

Answer

Before treating a client heading into head and neck radiation, chemotherapy or a stem cell transplant. Dental clearance, extraction timing, and management of mucositis and osteoradionecrosis risk are decided jointly with the oncologist, not scheduled independently by the dental office.

#154 · Collaboration

What causes conflict within a health care team?

Answer

Role ambiguity, competing priorities, unequal power, poor communication, workload pressure, and genuine differences in professional judgment. Some conflict is productive, because a team that never disagrees is usually a team where someone has stopped speaking.

#155 · Collaboration

Which conflict resolution approach suits a clinical disagreement affecting a client?

Answer

Collaborating, which looks for the interest behind each position rather than defending the position itself. The range of approaches is avoiding, accommodating, competing, compromising and collaborating. Competing is defensible only in an immediate safety emergency.

#156 · Collaboration

Where does conflict resolution start, and when do you escalate?

Answer

Start privately with the person involved, focused on the behaviour and the client rather than on personality. Escalate to a manager or the regulatory college only when that conversation fails or when the risk to the client is immediate.

#157 · Collaboration

How does a hygienist speak up about a safety concern?

Answer

Use graded escalation: state the observation, state the concern, state what you propose, and if the answer does not resolve it, stop and say clearly that you are not comfortable proceeding. Speaking up is a professional obligation, not an option.

#158 · Collaboration

Must a hygienist perform an act she believes will harm the client?

Answer

No. Every Canadian regulatory college expects a registrant to refuse to perform an act that would harm a client, regardless of who directed it. Following an instruction is not a defence for an act she knew or should have known was unsafe.

#159 · Collaboration

How is a dentist and hygienist disagreement about a treatment plan resolved?

Answer

Not by automatic deference and not by unilateral action. Discuss it privately with your evidence and reasoning, and bring the client in, because consent belongs to the client. If it continues and you believe the plan will harm the client, do not perform it and document the disagreement factually.

#160 · Collaboration

What are the conditions for safe delegation, and who stays accountable?

Answer

Delegate only when the task is within your own authorized scope, the person is competent and authorized to receive it, supervision is available and the client is stable. Accountability for the decision to delegate stays with you, and anyone not competent to perform a delegated act declines it.

#161 · Practice Management

What are the six links in the chain of infection?

Answer

Agent, reservoir, portal of exit, mode of transmission, portal of entry and susceptible host. Transmission needs all six links, so breaking any single one stops it. Sterilization attacks the agent, hand hygiene and barriers attack transmission, and immunization protects the susceptible host.

#162 · Practice Management

What does routine practices mean in Canadian infection prevention and control?

Answer

Routine practices are the Public Health Agency of Canada measures applied to every client, every time, regardless of diagnosis or presumed infectious status. Blood, saliva and all body fluids except sweat are treated as potentially infectious. The American equivalent term is standard precautions.

#163 · Practice Management

When are additional precautions used, and what do they add?

Answer

They are layered onto routine practices for a suspected organism. Contact precautions add gown and gloves, droplet precautions add mask and eye protection within two metres, and airborne precautions need a fit tested N95 and negative pressure. A dental operatory cannot provide negative pressure, so suspected tuberculosis is deferred and referred.

#164 · Practice Management

What are the four moments of hand hygiene?

Answer

Before initial client contact, before an aseptic procedure, after a body fluid exposure risk, and after contact with the client or the client environment. Hand hygiene is the single most effective preventive measure, and it is performed before donning gloves and immediately after removing them.

#165 · Practice Management

How is alcohol based hand rub used correctly?

Answer

Use a rub containing 60 to 90 percent alcohol when hands are not visibly soiled. Apply to all hand surfaces and rub for 20 to 30 seconds until dry, without wiping or fanning, because the alcohol needs its full contact time to work.

#166 · Practice Management

When must soap and water be used instead of alcohol rub?

Answer

When hands are visibly soiled, after using the washroom, and when exposure to spore forming organisms such as Clostridioides difficile or non-enveloped viruses such as norovirus is suspected, because alcohol does not kill spores. Lather 15 to 20 seconds, rinse with fingertips down, dry with a single use towel.

#167 · Practice Management

What is the correct order for donning personal protective equipment?

Answer

Hand hygiene first, then gown, then mask, then eye protection, and gloves last. Gloves go on last so that they cover the gown cuffs. This order is not simply reversed when the equipment comes off again.

#168 · Practice Management

What is the correct order for doffing personal protective equipment?

Answer

Gloves first because they are the most contaminated, then the gown rolled away from the body, then hand hygiene, then eye protection by its arms or headband, then the mask last by its ties or ear loops, then hand hygiene again. Doffing follows contamination, not the reverse of donning.

#169 · Practice Management

What is the correct sequence for instrument reprocessing?

Answer

Transport in a rigid closed puncture resistant container, clean, rinse and dry, inspect under good light, package and label with the sterilizer, load number and date, sterilize, then store clean, dry and closed. Flow runs one way from dirty to clean with no backtracking.

#170 · Practice Management

Why must cleaning always come before sterilization?

Answer

Blood, saliva and debris physically shield microorganisms from steam or heat, and dried bioburden insulates them further. A soiled instrument can therefore leave a perfect sterilization cycle still contaminated. Cleaning removes that bioburden, so it is never optional and never left until afterwards.

#171 · Practice Management

How does ultrasonic cleaning work, and how are instruments loaded?

Answer

It cleans by cavitation, where imploding bubbles lift debris from surfaces. Instruments are fully immersed, hinged instruments are opened, and nothing is stacked, or cleaning fails. Ultrasonic cleaners and instrument washers are preferred over hand scrubbing because they reduce handling of contaminated sharps.

#172 · Practice Management

What are the parameters for steam sterilization?

Answer

Gravity displacement runs about 121 degrees C at roughly 103 kPa for 15 to 30 minutes, and dynamic air removal about 132 to 135 degrees C for 3 to 4 minutes. Steam corrodes carbon steel, dulls some cutting edges and destroys heat sensitive plastics.

#173 · Practice Management

What are the parameters and uses of dry heat sterilization?

Answer

About 160 degrees C for 120 minutes, or 170 degrees C for 60 minutes, with rapid transfer units near 190 degrees C for 6 to 12 minutes. Timing starts only once the chamber reaches temperature. It suits carbon steel burs and orthodontic pliers but damages plastics and rubber.

#174 · Practice Management

What is unsaturated chemical vapour sterilization, and how are handpieces handled?

Answer

An alcohol and formaldehyde solution at about 132 degrees C and roughly 138 kPa for 20 minutes. Carbon steel does not rust, but items must be dry before loading and the room needs ventilation. Handpieces are heat sterilized between clients, and surface disinfecting a handpiece is never acceptable.

#175 · Practice Management

What is mechanical monitoring of a sterilizer, and what does it prove?

Answer

Mechanical monitoring is the gauges and printouts of time, temperature and pressure, recorded for every cycle. It shows only what the machine registered at its own sensor, not what conditions were reached inside a package, so it cannot confirm sterility on its own.

#176 · Practice Management

What do chemical indicators prove, and what are their classes?

Answer

They change colour on exposure to one or more cycle conditions, so they prove exposure and not sterility. ISO 11140 defines six classes, from Class 1 process indicators such as autoclave tape through Class 5 integrating and Class 6 emulating indicators. An internal indicator goes in every package.

#177 · Practice Management

What is a biological indicator, and why is it the only proof of sterilization?

Answer

A spore test carrying live, highly resistant spores, Geobacillus stearothermophilus for steam and chemical vapour and Bacillus atrophaeus for dry heat. Only a biological indicator proves sterilization was achieved, because only it challenges the cycle with living organisms more resistant than any pathogen present. Run it weekly and with every implantable load.

#178 · Practice Management

What is done when a spore test fails?

Answer

Remove the sterilizer from service and repeat with a fresh biological indicator. If a clear operator error such as overloading is found and the repeat is negative, it may return to service. Otherwise keep it out until serviced with three consecutive negative tests. Recall and reprocess every load back to the last negative test.

#179 · Practice Management

How does the Spaulding classification sort dental items?

Answer

Critical items penetrate soft tissue or bone, such as scalers, curettes, probes and blades, and are heat sterilized or single use. Semi critical items contact mucous membranes without penetrating, such as mirrors and handpieces, and are heat sterilized where possible. Non-critical items touch intact skin only and need cleaning plus low or intermediate level disinfection.

#180 · Practice Management

How are surface barriers and surface disinfection managed between clients?

Answer

Barriers are single use coverings changed between clients while gloved, and an intact barrier means the surface underneath needs no disinfection until day end. Otherwise clean first, then disinfect with a product carrying a Drug Identification Number, keeping the surface visibly wet for the full stated contact time, commonly one to ten minutes.

#181 · Practice Management

Why are dental unit waterlines a risk, and how are they controlled?

Answer

Narrow bore tubing and long stagnant periods let biofilm form and shed heterotrophic bacteria, Legionella and Pseudomonas. Non-surgical water must meet 500 colony forming units per millilitre or fewer. Use treatment products, independent reservoirs, filters, shocking and testing, flush 20 to 30 seconds after each client, and use sterile irrigant for surgery.

#182 · Practice Management

What are the rules for safe sharps handling?

Answer

Never recap two handed, and use a one handed scoop technique or a mechanical recapping device. Do not bend, break or shear needles, and pass instruments through a neutral zone. Place sharps in a puncture resistant labelled container at the point of use and replace it at the fill line.

#183 · Practice Management

What are the immediate steps after a percutaneous exposure?

Answer

Stop, remove gloves and wash the site with soap and water. Let it bleed gently, without squeezing it or applying caustic agents. Flush exposed eyes or mucous membranes with water or saline for about 15 minutes. Report immediately and complete an incident report.

#184 · Practice Management

What does post exposure medical assessment cover?

Answer

Seek assessment urgently, ideally within one to two hours. It reviews the exposure type, the worker immune status, and the source hepatitis B, hepatitis C and HIV status. HIV prophylaxis is not offered beyond 72 hours. A non-immune worker exposed to hepatitis B receives immune globulin plus vaccine, and hepatitis C has no prophylaxis.

#185 · Practice Management

How are aerosols controlled during ultrasonic scaling?

Answer

High volume evacuation with a wide bore tip held close to the working end is the most effective control, and a saliva ejector does not achieve this. A pre-procedural rinse such as 0.12 percent chlorhexidine gluconate lowers the oral microbial load. Ventilation, settle time and a fit tested respirator help further.

#186 · Practice Management

Which immunizations are recommended for dental personnel?

Answer

Hepatitis B with post-vaccination anti-HBs testing to confirm protection, annual influenza vaccine, measles mumps and rubella, varicella, and tetanus diphtheria and pertussis. Hepatitis B matters most because it is the only bloodborne pathogen met in dentistry that has a vaccine.

#187 · Practice Management

Why is the client record described as a legal document?

Answer

It is the practitioner's main defence in a complaint or civil claim, and care that was not documented is treated as care that was not provided. It must hold the updated health history, vital signs, extraoral and intraoral findings, charting, radiographs with justification, the dental hygiene diagnosis and the care plan.

#188 · Practice Management

How is a treatment entry structured, and what must it record?

Answer

Use subjective findings, objective findings, assessment or dental hygiene diagnosis, and plan. Then record what was actually done: areas treated, agents used, anesthetic type, dose and lot number, the client's response, instruction given, the recall interval, and any cancellations, no-shows or referrals. Charting uses FDI two digit notation.

#189 · Practice Management

What must be documented for informed consent and informed refusal?

Answer

Consent is recorded, not assumed: the proposed care, the risks, the alternatives and the consequences of no treatment. Informed refusal is recorded the same way, naming what was recommended, the risks of declining, and the client's decision.

#190 · Practice Management

How long are client records retained?

Answer

Retention is set provincially, commonly ten years from the last entry for an adult and, for a minor, ten years from the age of majority. A record connected to a complaint or a claim is never destroyed, whatever the usual retention period allows.

#191 · Practice Management

How is an error in a client record corrected?

Answer

Draw a single line through it so the original stays legible, add the correct information, then date and initial it. Never erase, white out or delete. A late entry is labelled as such and names the date of the event. Altering a record afterwards is professional misconduct.

#192 · Practice Management

What extra safeguards do electronic records require?

Answer

Unique logins, role based access, automatic screen lock, encryption and secure backup, plus an audit trail recording who viewed or changed what and when. That audit trail is why an electronic entry cannot simply be overwritten, since a correction is added rather than substituted.

#193 · Practice Management

Which privacy laws govern client health information in Canada?

Answer

PIPEDA applies federally, alongside provincial statutes such as PHIPA in Ontario, the Health Information Act in Alberta and PIPA in British Columbia. Collect only the information needed and disclose it only with consent or where the law requires disclosure.

#194 · Practice Management

What is the correct order of a privacy breach response?

Answer

Contain the breach first, evaluate the risk, notify the privacy officer, then the affected individuals and the privacy commissioner and college where required, and finally act to prevent recurrence. Containment comes before notification because unchecked exposure keeps widening the harm.

#195 · Practice Management

How is a recall interval decided?

Answer

Recall is risk based, not automatic. Treated periodontitis, poor plaque control, diabetes, xerostomia or a smoking history may justify three month periodontal maintenance, while a low risk client with stable tissues may safely be seen at nine to twelve months. Document the reasoning.

#196 · Practice Management

Which billing practices count as fraud?

Answer

Billing for services not rendered, changing a procedure code so a non-covered service is reimbursed, and reporting a false treatment date to fit a benefit year. Waiving a co-payment while billing the insurer the full fee misrepresents the actual fee. All of these are professional misconduct.

#197 · Practice Management

What must a WHMIS 2015 supplier label and safety data sheet contain?

Answer

A supplier label carries the product identifier, initial supplier identifier, pictogram, the signal word Danger or Warning, hazard statements and precautionary statements. A workplace label on a decanted container names the product, safe handling and the safety data sheet. Sheets use 16 sections and stay accessible every shift.

#198 · Practice Management

How does a hygienist reduce musculoskeletal injury risk?

Answer

Work seated in neutral posture with the back supported, hips at or slightly above ninety degrees and feet flat. Position the client supine and rotate their head rather than bending your own neck. Use loupes and indirect vision, keep instruments sharp, choose lightweight larger diameter handles and take stretch breaks.

#199 · Practice Management

What is the employer duty for radiation safety?

Answer

Provincial radiation legislation and Health Canada's Safety Code 30 require registering and maintaining equipment, appointing a responsible user, keeping a written radiation protection plan, providing shielding and dosimeters where required, and ensuring operators are qualified. Clinical work follows ALARA, and no staff member ever holds a receptor during exposure.

#200 · Practice Management

What is the purpose of an incident report?

Answer

It is an internal risk management document covering needlesticks, medication errors, falls and privacy breaches. Facts are recorded objectively without blame, then the root cause is analyzed and acted on. A clinical incident is also documented factually in the client chart, which stays separate from the report.

#201 · Prevention Education and Health Promotion

What is dental biofilm?

Answer

An organized community of micro-organisms attached to a tooth surface within a matrix of bacterial and salivary polymers. It is the primary etiological factor in both caries and gingivitis. Gingivitis resolves when biofilm is removed and returns when it reaccumulates, while in caries removal halts the process and a non cavitated lesion can remineralize, but a cavitated lesion does not reverse.

#202 · Prevention Education and Health Promotion

Describe the acquired pellicle and its role in biofilm formation.

Answer

Salivary glycoproteins, phosphoproteins and enzymes adsorb onto cleaned enamel within seconds. The pellicle is acellular and bacteria free when it forms, cannot be rinsed away, and supplies the receptors that bacteria bind to, so it is the first stage of biofilm formation.

#203 · Prevention Education and Health Promotion

Outline the timeline of biofilm maturation.

Answer

Early colonizers attach within hours, mostly gram positive cocci and rods such as Streptococcus sanguinis and Actinomyces. Between one and three days filaments and gram negative anaerobes join by coaggregation. By seven to fourteen days the biofilm is mature and gingival inflammation is visible.

#204 · Prevention Education and Health Promotion

Why must biofilm be disrupted mechanically rather than treated with a rinse alone?

Answer

Bacteria inside an organized biofilm resist antimicrobials far better than the same species free in saliva, and a rinse cannot penetrate an established mass in thirty seconds. Breaking the structure apart resets the community to its early health associated state, so every chemotherapeutic agent is an adjunct.

#205 · Prevention Education and Health Promotion

Explain demineralization and remineralization in the caries process.

Answer

Bacteria ferment carbohydrate into organic acids, mainly lactic acid, and plaque pH falls below the critical pH, so calcium and phosphate leave the tooth. When the challenge ends saliva buffers the acid, pH rises and mineral returns. A lesion forms only when demineralization outweighs remineralization over time.

#206 · Prevention Education and Health Promotion

What is the critical pH for enamel?

Answer

Approximately 5.5. Below this value the plaque fluid is undersaturated with respect to enamel mineral, so calcium and phosphate leave the tooth surface. This number applies to enamel only and is different from the critical pH of exposed root surface.

#207 · Prevention Education and Health Promotion

What is the critical pH for root surface dentin and cementum, and why does it matter?

Answer

Approximately 6.2 to 6.7, which is higher and therefore less acidic than the enamel value of about 5.5. Root surfaces lose mineral at a pH that intact enamel tolerates safely, so a client with recession is vulnerable to acid exposures that spare the crown.

#208 · Prevention Education and Health Promotion

Contrast Streptococcus mutans and Lactobacillus species in caries.

Answer

Streptococcus mutans is most strongly associated with lesion initiation because it adheres well to pellicle, makes extracellular polysaccharide from sucrose, and is both acidogenic and acid tolerant. Lactobacillus adheres poorly to smooth surfaces and drives progression, colonizing cavitated lesions and deep fissures.

#209 · Prevention Education and Health Promotion

What does the Stephan curve show?

Answer

Plaque pH after a carbohydrate exposure. Resting pH sits near 6.5 to 7.0. After a sucrose rinse pH drops sharply within two to five minutes, dips below the critical pH, then recovers over twenty to sixty minutes. The area below the critical pH line is the demineralization window.

#210 · Prevention Education and Health Promotion

Why does frequency of sugar exposure matter more than the amount?

Answer

Each exposure produces its own Stephan curve. A can of pop drunk in ten minutes produces one curve, while the same can sipped across four hours produces overlapping curves and far more time below the critical pH. A bedtime exposure is worst, because salivary flow falls close to zero during sleep.

#211 · Prevention Education and Health Promotion

What is the purpose of caries risk assessment?

Answer

It places a client at low, moderate or high risk. That category then drives the recall interval, the fluoride regimen and the depth of diet counselling, so the assessment converts findings into an individualized preventive plan rather than generic advice.

#212 · Prevention Education and Health Promotion

List indicators that place a client at high caries risk.

Answer

Any new lesion within twelve months, white spot lesions, exposed root surfaces, heavy biofilm, frequent snacking or sipping, reduced salivary flow, deep fissures, orthodontic appliances, low fluoride exposure and low income.

#213 · Prevention Education and Health Promotion

Define early childhood caries and describe its classic pattern and drivers.

Answer

Caries in any primary tooth in a child under six. The maxillary primary incisors are affected first while the mandibular incisors are protected by the tongue and sublingual ducts. Drivers are frequent sweetened liquids, sleeping with a bottle, nocturnal breastfeeding after teeth erupt, and flora acquired from a caregiver.

#214 · Prevention Education and Health Promotion

Why does root caries dominate in older adults?

Answer

Recession exposes cementum and dentin, which demineralize at a higher critical pH of about 6.2 to 6.7 than enamel. Dry mouth from polypharmacy, reduced dexterity and a softer more frequently eaten diet add to that exposure, so risk rises sharply.

#215 · Prevention Education and Health Promotion

Why is xerostomia among the strongest caries drivers?

Answer

Saliva provides clearance of food and acid, buffering capacity, a calcium and phosphate reservoir for remineralization, and antimicrobial proteins. Losing it removes all four protections at once, so rampant cervical caries in a client with a long medication list calls for a salivary assessment.

#216 · Prevention Education and Health Promotion

Name common causes of xerostomia.

Answer

Anticholinergics, antidepressants, antihypertensives, diuretics and antihistamines, along with head and neck radiation and Sjogren syndrome. Polypharmacy in older adults makes medication the most common cause seen in practice.

#217 · Prevention Education and Health Promotion

Does fluoride work mainly systemically or topically?

Answer

Mainly by topical mechanisms after eruption, not mainly by being swallowed during tooth formation. This is the most commonly tested concept in the topic, and it is why regular post-eruptive exposure from toothpaste, fluoridated water and professional varnish outweighs any pre-eruptive systemic effect.

#218 · Prevention Education and Health Promotion

By what mechanisms does fluoride reduce caries?

Answer

Fluoride in the fluid around the tooth inhibits demineralization during an acid challenge, accelerates remineralization, and drives formation of fluorapatite, which dissolves at a lower pH than hydroxyapatite. At higher concentrations it also inhibits bacterial enolase and reduces acid production.

#219 · Prevention Education and Health Promotion

State the Canadian water fluoridation concentrations and why fluoridation is valued.

Answer

Health Canada sets the optimal concentration for communities that fluoridate at 0.7 mg/L. The maximum acceptable concentration for drinking water safety is 1.5 mg/L, a ceiling rather than a target. Fluoridation is the most equitable preventive measure because it reaches everyone regardless of income or dental attendance.

#220 · Prevention Education and Health Promotion

What fluoride concentrations are found in Canadian toothpastes?

Answer

Toothpaste sold for general use contains roughly 1000 to 1500 parts per million fluoride. Prescription paste at 5000 ppm suits high risk adults, particularly those with root caries or xerostomia.

#221 · Prevention Education and Health Promotion

How much toothpaste is used from birth to three years?

Answer

An adult brushes the child's teeth with a rice grain sized smear of fluoridated toothpaste, and only when the child has been assessed as at risk. The small amount limits swallowing while still delivering topical fluoride to erupted surfaces.

#222 · Prevention Education and Health Promotion

How much toothpaste is used from three to six years?

Answer

A green pea sized amount of fluoridated toothpaste, used with adult supervision, and the child is taught to spit rather than swallow. This is larger than the rice grain smear used before three years.

#223 · Prevention Education and Health Promotion

What is the fluoride concentration of professional varnish, and why suit children?

Answer

Five percent sodium fluoride varnish contains 22 600 ppm fluoride. It sets on contact with saliva and so little material is used that swallowing risk is low, which makes it the professional agent of choice for children.

#224 · Prevention Education and Health Promotion

Give the application and aftercare instructions for fluoride varnish.

Answer

Dry the teeth and apply a thin layer to all surfaces. Afterwards the client takes soft food only, avoids hot drinks and alcohol containing rinses that day, and does not brush or floss until the next morning, or for at least four to six hours. High risk clients are treated two to four times yearly.

#225 · Prevention Education and Health Promotion

Why are fluoride drops and tablets rarely recommended now?

Answer

They act systemically when the benefit is largely topical, they carry a fluorosis risk, and compliance is poor. Where considered at all, they are only for a high risk child without a fluoridated water supply, and never under six months of age.

#226 · Prevention Education and Health Promotion

What is dental fluorosis, when can it occur, and from what sources?

Answer

An enamel defect from excess fluoride intake while enamel is forming, ranging from white flecking to brown staining with pitting. The window runs from birth to about six to eight years. Usual sources are swallowed toothpaste, supplements and fluoridated water used for infant formula. An erupted tooth cannot become fluorotic.

#227 · Prevention Education and Health Promotion

What is the probable toxic dose of fluoride and its early signs?

Answer

Five milligrams of fluoride per kilogram of body weight, at or above which intervention and hospital assessment are indicated. Early signs are nausea, vomiting, abdominal pain and hypersalivation. Severe poisoning causes hypocalcemia, arrhythmia and death.

#228 · Prevention Education and Health Promotion

Outline first aid for acute fluoride ingestion by dose band.

Answer

Below 5 mg/kg, give milk or another calcium source and observe. Between 5 and 15 mg/kg, give calcium by mouth, induce vomiting and transport to hospital. Above 15 mg/kg, transport immediately with cardiac monitoring.

#229 · Prevention Education and Health Promotion

What does silver diamine fluoride do, and what are its drawback and indications?

Answer

Silver diamine fluoride at 38 percent is painted on a carious lesion to arrest it. Silver is antibacterial and fluoride remineralizes the surface, producing hard arrested dentin without drilling. Arrested dentin turns black permanently, though sound enamel is not stained. It suits root caries in frail older adults and limited access to care.

#230 · Prevention Education and Health Promotion

How do powered and manual toothbrushes compare?

Answer

Both remove biofilm effectively when used properly, though systematic review evidence shows a small advantage for powered brushes, particularly oscillating rotating designs. Powered brushes suit clients with reduced dexterity or orthodontic appliances.

#231 · Prevention Education and Health Promotion

Describe the Bass technique and its main indication.

Answer

A soft multi tufted brush is placed at a 45 degree angle to the long axis of the tooth, bristle tips directed apically into the gingival sulcus. Short vibratory strokes with light pressure clean two or three teeth at a time. Targeting the sulcus makes it the technique of choice for gingivitis.

#232 · Prevention Education and Health Promotion

State the recommended toothbrushing regimen.

Answer

Brush for two minutes twice daily with fluoridated toothpaste, then spit without vigorous rinsing so fluoride stays at the tooth surface. Replace the brush every three to four months, or sooner if the bristles splay.

#233 · Prevention Education and Health Promotion

How does toothbrush abrasion present and how is it corrected?

Answer

Smooth wedge shaped cervical defects on facial surfaces of canines and premolars, worse on the side opposite the dominant hand. It follows horizontal scrubbing, heavy force, hard bristles or an abrasive dentifrice. The correction is a change of technique, not only a change of brush.

#234 · Prevention Education and Health Promotion

Interdental brush or floss? What does current evidence favour?

Answer

Current evidence favours interdental brushes over floss for reducing plaque and gingival inflammation wherever the space allows a brush to fit passively. Floss remains the aid of choice for tight contacts where the papilla fills the embrasure, since forcing a brush there traumatizes the tissue.

#235 · Prevention Education and Health Promotion

Match the interdental aid to each embrasure type.

Answer

Type I, the papilla fills the space, so floss is indicated. Type II, the papilla is partly lost, so a small interdental brush, soft wood stick or rubber tip suits. Type III, the papilla is gone, so a larger interdental brush or an end tuft brush is needed.

#236 · Prevention Education and Health Promotion

How is floss used correctly, and what helps a client with poor dexterity?

Answer

Floss is curved into a C shape against each proximal surface and moved under the papilla, not snapped through the contact. Floss holders and pre threaded flossers help clients with poor dexterity, or a caregiver cleaning another person's mouth. Wood sticks suit facial access only.

#237 · Prevention Education and Health Promotion

What place do oral irrigators have in home care?

Answer

They reduce bleeding and reach pockets and fixed appliances well, but they flush rather than disrupt adherent biofilm. For that reason an irrigator is prescribed alongside brushing and interdental cleaning, never as a replacement for them.

#238 · Prevention Education and Health Promotion

Give the regimen and indications for chlorhexidine gluconate rinse.

Answer

Chlorhexidine at 0.12 percent is the most substantive antiplaque agent in Canada, binding to oral surfaces and releasing slowly over eight to twelve hours. The usual regimen is 15 mL rinsed for 30 seconds twice daily. Indications are short term: after periodontal therapy, for necrotizing gingival conditions, and as a pre procedural rinse.

#239 · Prevention Education and Health Promotion

What are the side effects and use limits of chlorhexidine?

Answer

Extrinsic brown staining of teeth, restorations and tongue, altered or reduced taste, increased supragingival calculus, and mucosal desquamation. Use is usually about two weeks and rarely beyond two months. Anionic detergents in toothpaste inactivate it, so the rinse is used about 30 minutes apart from brushing.

#240 · Prevention Education and Health Promotion

Compare essential oil rinses and cetylpyridinium chloride with chlorhexidine.

Answer

Essential oil rinses containing thymol, eucalyptol, menthol and methyl salicylate reduce plaque and gingivitis without staining, making them better for long term daily use. Alcohol containing formulations are unsuitable for children, xerostomia or mucositis, and alcohol dependence recovery. Cetylpyridinium chloride at 0.05 to 0.075 percent has a modest effect and lower substantivity.

#241 · Prevention Education and Health Promotion

How should a client care for removable dentures?

Answer

Brush daily with a denture brush and mild soap or a denture cleanser, never abrasive toothpaste, over a basin part filled with water. Hot water warps acrylic and sodium hypochlorite corrodes cast metal frameworks. Dentures come out overnight and are stored in water, since nocturnal wear drives denture stomatitis caused by Candida albicans.

#242 · Prevention Education and Health Promotion

How is biofilm controlled around implants and fixed prostheses?

Answer

Use soft brushes, end tuft brushes, and interdental brushes with nylon coated wire, because an exposed metal core scratches the titanium abutment. Low abrasive paste and a low setting irrigator are appropriate. Around fixed bridges and orthodontic appliances, floss threaders and tufted floss carry cleaning material under the pontic or behind the archwire.

#243 · Prevention Education and Health Promotion

Define cariogenic, cariostatic and anticariogenic foods.

Answer

Cariogenic foods contain fermentable carbohydrate that bacteria convert to acid. Cariostatic foods are not metabolized to acid and do not lower plaque pH, including proteins, fats, most vegetables and sugar alcohols such as xylitol and sorbitol. Anticariogenic foods actively interfere with the process, aged cheese being the standard example.

#244 · Prevention Education and Health Promotion

How are hidden sugars and sports drinks addressed in diet counselling?

Answer

Sucrose, glucose, fructose, maltose, dextrose, honey, molasses and concentrated fruit juice all behave the same way in plaque, so teaching a client to scan the ingredient list for words ending in ose beats a list of forbidden foods. Sports and energy drinks add acidity near pH 3 and are sipped slowly.

#245 · Prevention Education and Health Promotion

How is a diet history taken and analyzed?

Answer

Take the history before advising. A 24 hour recall screens quickly, while a three to five day food diary including one non working day gives a better picture. Analyze it for the number, timing and form of fermentable carbohydrate exposures, then feed those findings back to the client.

#246 · Prevention Education and Health Promotion

How does erosion differ from caries?

Answer

Erosion is chemical dissolution of tooth surface by acid of non bacterial origin, so neither biofilm nor fermentable carbohydrate is required. It produces smooth, shiny, cupped lesions in a generalized distribution, whereas caries produces localized opaque or cavitated lesions at plaque retentive sites.

#247 · Prevention Education and Health Promotion

Name the sources of dental erosion and the advice given.

Answer

Extrinsic sources include citrus, carbonated and sports drinks and wine. Intrinsic sources include reflux and vomiting, which classically affects the palatal surfaces of the maxillary anterior teeth. Advise rinsing with water or a bicarbonate solution afterwards, delaying brushing 30 to 60 minutes, using a straw, and using fluoride and sugar free gum.

#248 · Prevention Education and Health Promotion

What makes an oral health behaviour change goal effective?

Answer

It is specific and achievable, naming what to do, where and when, such as using the interdental brush between the lower right molars every night before bed for three weeks. Set one or two goals per visit, starting with one the client is confident of achieving, and add self monitoring such as a calendar tick.

#249 · Prevention Education and Health Promotion

What oral effects follow tobacco and vaping use?

Answer

Tobacco causes staining, halitosis, smoker's melanosis, nicotine stomatitis, leukoplakia and oral cancer, with greater attachment and bone loss, poorer therapy response, higher implant failure and delayed healing. Nicotine vasoconstriction suppresses bleeding on probing, so the gingiva can look deceptively healthy. Vaping is linked to dry mouth, mucosal irritation, gingival inflammation and caries from sweetened aerosols.

#250 · Prevention Education and Health Promotion

Describe the five A framework for brief cessation intervention.

Answer

Ask every client about use and record it. Advise quitting clearly, linked to a finding in their own mouth. Assess willingness to make an attempt now. Assist the willing with a quit date, counselling and information about pharmacotherapy. Arrange follow up, or refer to a provincial quitline.

#251 · Prevention Education and Health Promotion

What are the social determinants of health relevant to oral disease?

Answer

The conditions people live in rather than their biology or habits: income, education and literacy, employment and working conditions, housing, food security, social exclusion, geography, Indigenous status, disability, and experiences of racism and discrimination. Each one has a direct pathway to oral health.

#252 · Prevention Education and Health Promotion

How does income act as a determinant of oral health in Canada?

Answer

Income decides whether a client can pay for a scaling, a filling or a denture, and cost is the reason Canadians most often give for avoiding dental care. Because dental care is financed largely privately, ability to pay rather than clinical need governs who receives treatment.

#253 · Prevention Education and Health Promotion

Why does employment matter twice for oral health?

Answer

Work supplies the money to pay for care, and most private dental insurance in Canada is obtained through a job. Part time, contract and gig work therefore leave the people with the least income also holding the least coverage, which compounds the barrier rather than offsetting it.

#254 · Prevention Education and Health Promotion

How do housing and food insecurity affect oral health?

Answer

Brushing twice daily assumes a sink, storage and privacy, none of which are guaranteed in unstable or crowded housing. Food insecurity raises caries risk because cheap, shelf stable food is often high in fermentable carbohydrate, so the diet reflects budget and access rather than preference.

#255 · Prevention Education and Health Promotion

What is the social gradient in oral disease, and what causes the Indigenous burden?

Answer

Oral disease rises steadily as income falls, producing a gradient across the whole population rather than a simple poor versus rich split. Colonization and residential schools are why Indigenous populations carry a caries burden far above the national average. Racism and discrimination act as determinants in their own right.

#256 · Prevention Education and Health Promotion

How does the Canada Health Act treat dental care?

Answer

The Act sets the conditions provinces must meet to receive federal health transfers and covers medically necessary hospital and physician services. Dental care is largely outside it. Routine examinations, scaling, restorations, office extractions, dentures and orthodontics are not insured services under the Act.

#257 · Prevention Education and Health Promotion

What dental work does fall inside the Canada Health Act?

Answer

Only the narrow exception of surgical dental procedures that must be performed in a hospital setting, such as some maxillofacial surgery, or treatment under general anesthesia for a client whose medical condition requires a hospital. The setting and medical necessity, not the tooth, decide coverage.

#258 · Prevention Education and Health Promotion

How is dental care financed in Canada, and why does that matter?

Answer

Mostly privately, through employer sponsored insurance and out of pocket payment, with public plans covering only part of the population. Coverage therefore follows employment and income rather than need, which is the structural reason oral disease is socially patterned in a country with public medical insurance.

#259 · Prevention Education and Health Promotion

What do provincial and territorial dental programs typically cover?

Answer

They differ by jurisdiction in who qualifies and what is included. Common patterns are programs for children in lower income households, programs tied to social assistance and disability income support, and programs for some seniors. Because they often pay below the provincial fee guide, some offices decline to bill them.

#260 · Prevention Education and Health Promotion

Describe the structure of the federal Canadian Dental Care Plan.

Answer

Health Canada administers it with a benefits administrator handling claims. It is income tested, targets residents without access to private dental insurance, requires a filed tax return, may involve a copayment, does not replace provincial or private plans, and leaves participation voluntary for providers.

#261 · Prevention Education and Health Promotion

Why should you avoid quoting Canadian Dental Care Plan eligibility details from memory?

Answer

Enrolment has been phased in by group over several years, and eligibility ages, income ceilings and covered procedures changed during the phase in and can change again. Describe the structure only, and direct clients to the current official source rather than stating a threshold as permanent fact.

#262 · Prevention Education and Health Promotion

What is the Non Insured Health Benefits program?

Answer

A federal program administered by Indigenous Services Canada for registered First Nations people and recognized Inuit. It covers a defined range of dental services along with vision care and medical transportation. It is a benefits program, not a provincial plan, and it does not cover every service a client may need.

#263 · Prevention Education and Health Promotion

What are two known limitations of Non Insured Health Benefits coverage?

Answer

Many services require predetermination, meaning the provider submits a treatment plan for approval before care proceeds, which delays treatment. Metis and non status First Nations people are generally not covered at all. Cultural safety and genuine community partnership remain conditions for any program here to work.

#264 · Prevention Education and Health Promotion

What barriers to dental care do newcomers and refugees face?

Answer

Language, unfamiliarity with a fee for service system, work without benefits after non recognition of foreign credentials, and fear of cost. Government assisted refugees and protected persons may hold temporary federal coverage for limited urgent dental care, which ends once other coverage begins.

#265 · Prevention Education and Health Promotion

What service model works for people experiencing homelessness?

Answer

Untreated caries, periodontitis and tooth loss are all very high in this group, alongside substance use, smoking and xerostomia. Fixed appointment models fail. Outreach into shelters and drop in centres that accepts walk ins and requires no fixed address reaches these clients far more effectively.

#266 · Prevention Education and Health Promotion

What barriers to care do people with disabilities encounter?

Answer

Operatory access, transfer equipment, appointment length, sedation availability, dependence on a caregiver, and provider discomfort. Adults with intellectual and developmental disabilities carry more untreated disease than the general population, so the barrier is largely in how services are designed rather than in the client.

#267 · Prevention Education and Health Promotion

What is the core access problem in rural and remote communities?

Answer

Workforce distribution rather than absolute numbers, since many rural communities have no resident hygienist or dentist. Responses include portable equipment, teledentistry and rotating clinics. Independent or self initiated dental hygiene practice, available in several provinces, exists partly to serve these gaps.

#268 · Prevention Education and Health Promotion

Why is oral health poor in long term care, and what fixes it?

Answer

Residents keep more natural teeth than earlier generations, take multiple xerostomic medications and depend on staff who often lack training and time. Rampant root caries, denture stomatitis and aspiration pneumonia follow. Effective responses are structural: an oral care policy, assessment on admission, bedside supplies, staff training and an attached hygienist.

#269 · Prevention Education and Health Promotion

Define health education and give an oral health example.

Answer

Health education is planned learning activity that improves knowledge and skills so people can make informed choices. A toothbrushing demonstration is health education. It works on the individual's capability and does not by itself change the policies or environments that shape whether the behaviour is possible.

#270 · Prevention Education and Health Promotion

Define disease prevention as distinct from health promotion.

Answer

Disease prevention reduces the occurrence or progression of one specific disease, so placing sealants prevents caries. It is disease focused and usually delivered as a defined intervention, whereas health promotion is broader and addresses policy, environments and community power rather than a single condition.

#271 · Prevention Education and Health Promotion

How does the World Health Organization define health promotion?

Answer

The process of enabling people to increase control over and improve their health. It is broader than education or prevention because it adds policy, environments and community power. Removing sugary drinks from a school is health promotion, while teaching students about sugar is health education.

#272 · Prevention Education and Health Promotion

What is the Ottawa Charter and what three strategies does it name?

Answer

The standard health promotion framework, adopted in 1986 at a World Health Organization conference in Ottawa. Its three core strategies are advocate, enable and mediate. It also sets out five action areas that describe where health promotion work actually happens.

#273 · Prevention Education and Health Promotion

Name the five action areas of the Ottawa Charter.

Answer

Build healthy public policy, create supportive environments, strengthen community action, develop personal skills, and reorient health services. Together they move health promotion beyond individual instruction toward the policies, settings, communities and service systems that determine whether healthy choices are available.

#274 · Prevention Education and Health Promotion

Give an oral health example of each Ottawa Charter action area.

Answer

Healthy public policy: a fluoridation bylaw or a required oral care policy in long term care. Supportive environments: safe water in schools, shelters stocking toothbrushes. Strengthen community action: working through a community advisory committee. Develop personal skills: health education. Reorient health services: shifting the system from treatment toward prevention.

#275 · Prevention Education and Health Promotion

What is primary prevention, with dental examples?

Answer

Action taken before disease occurs, to stop it arising at all. Dental examples include community water fluoridation, pit and fissure sealants, mouthguards to prevent traumatic injury, and tobacco use prevention. The target population is people who do not yet have the disease.

#276 · Prevention Education and Health Promotion

What is secondary prevention, with dental examples?

Answer

Detecting and treating disease early to limit damage before it becomes advanced. Dental examples include screening, radiographic caries detection, remineralizing an early enamel lesion with fluoride varnish, and scaling for early gingivitis. Disease is already present but is still reversible or easily contained.

#277 · Prevention Education and Health Promotion

What is tertiary prevention, with dental examples?

Answer

Limiting disability and restoring function once disease is established and damage has occurred. Dental examples include complex restorative care, dentures, periodontal maintenance therapy, and rehabilitation after oral cancer surgery. The aim shifts from stopping disease to managing its consequences and restoring the client's function.

#278 · Prevention Education and Health Promotion

Explain the upstream, midstream and downstream metaphor with dental examples.

Answer

The metaphor describes people being pulled from a river. Downstream work rescues those already drowning, which is treatment such as restoring a carious tooth. Midstream work changes individual behaviour, such as diet counselling. Upstream work asks who is pushing people in, meaning policy and social conditions, such as fluoridation.

#279 · Prevention Education and Health Promotion

What is Rose's high risk strategy, and what are its limits?

Answer

It identifies the individuals at greatest risk and intervenes with them. It is efficient per person and fits clinical practice well. Its limits are that it needs a screening step, reaches only those actually identified, and leaves the underlying cause of the risk untouched in the wider population.

#280 · Prevention Education and Health Promotion

What is the population strategy and the prevention paradox?

Answer

The population strategy shifts the whole distribution of risk slightly for everyone. The prevention paradox is that most cases arise from the many at moderate risk rather than the few at high risk, so a small shift across everyone prevents more disease while offering little visible benefit to any one participant.

#281 · Prevention Education and Health Promotion

How do population and individual measures differ in their effect on inequity?

Answer

Population measures such as water fluoridation reach everyone regardless of income or attendance, so they narrow the social gradient. Measures depending on individual effort, purchase or compliance are taken up most by those with the most resources, so they tend to widen inequities even when they work.

#282 · Prevention Education and Health Promotion

What happens during needs assessment in community program planning?

Answer

Data are gathered before anything is decided, using surveillance data, screening surveys, sociodemographic profiles, an inventory of existing services, and consultation with the community. It is the first stage of the planning cycle, and it is also fed by the evaluation of the previous program.

#283 · Prevention Education and Health Promotion

Distinguish normative, felt, expressed and comparative need.

Answer

Normative need is defined by professionals against a standard. Felt need is what the community says it wants. Expressed need is felt need turned into demand, shown by people seeking services. Comparative need is judged by comparing one population against a similar population that is better served.

#284 · Prevention Education and Health Promotion

On what criteria are community needs prioritized?

Answer

Priority setting ranks identified needs against available resources using the size of the problem, its seriousness, the effectiveness and cost of the interventions available, and feasibility. It follows needs assessment, because you cannot rank problems responsibly before you have measured them.

#285 · Prevention Education and Health Promotion

How does a program goal differ from an objective?

Answer

A goal is a broad statement of intent, such as improving the oral health of school aged children in a region. An objective must be measurable and time limited, stating who, what, how much and by when, for example a stated percentage of grade two students with sealants by a named date.

#286 · Prevention Education and Health Promotion

Distinguish formative evaluation from process evaluation.

Answer

Formative evaluation happens while the program is still being developed, testing materials and methods before launch so they can be revised. Process evaluation asks whether the program was actually delivered as planned once it is running, counting sessions held, materials distributed and attendance.

#287 · Prevention Education and Health Promotion

What does impact evaluation measure?

Answer

The immediate effects of the program, measured soon after delivery. Examples include improved knowledge scores, the number of sealants placed, or lower plaque scores. It sits between process evaluation, which counts delivery, and outcome evaluation, which measures the long term change in health status.

#288 · Prevention Education and Health Promotion

What does outcome evaluation measure, and why run both process and outcome?

Answer

Outcome evaluation measures the long term change in health status the program existed to achieve, such as a reduced caries increment three years later. A program can pass process evaluation, having been delivered exactly as planned, and still fail outcome evaluation, so both are needed to interpret the result.

#289 · Prevention Education and Health Promotion

Distinguish prevalence from incidence in oral epidemiology.

Answer

Prevalence is the proportion of a population with a condition at one point in time and answers how much disease exists now. Incidence is the number of new cases arising over a defined period and answers how fast disease is appearing. A preventive program changes incidence.

#290 · Prevention Education and Health Promotion

What does the DMFT index count, and what is its maximum?

Answer

Caries experience in the permanent dentition. D is teeth with untreated decay, M is teeth missing because of caries, and F is teeth filled because of caries. The score is the sum at one point per tooth, to a maximum of 28 excluding third molars or 32 including them.

#291 · Prevention Education and Health Promotion

Why is DMFT a poor measure of current disease activity?

Answer

Each tooth is counted once only, in the most severe applicable category, and the index is cumulative and irreversible, so it can only rise across a lifetime and never fall. A client with a DMFT of 12 who is caries free today scores the same as one with twelve active lesions.

#292 · Prevention Education and Health Promotion

How does dmft differ from DMFT?

Answer

The dmft index is written in lower case and applies to the primary dentition. The critical difference is the m component, since primary teeth exfoliate naturally, so one is counted missing only when extracted because of caries and only at an age where exfoliation is not expected. Many surveys report dft instead.

#293 · Prevention Education and Health Promotion

Describe the Community Periodontal Index and its main limitation.

Answer

Six sextants are examined at index teeth with a WHO probe. Codes: 0 healthy, 1 bleeding, 2 calculus or defective margin, 3 pocket of 4 to 5 mm, 4 pocket of 6 mm or more. The highest code scores the sextant. It records only the worst site and cannot describe attachment loss.

#294 · Prevention Education and Health Promotion

Which oral health indices are reversible, and why does that matter?

Answer

The Gingival Index of Loe and Silness, the Plaque Index of Silness and Loe, and the Simplified Oral Hygiene Index of Greene and Vermillion, which runs 0 to 6 as debris plus calculus. Reversible indices improve after treatment, so they can evaluate a program, unlike the cumulative DMFT.

#295 · Prevention Education and Health Promotion

How does screening differ from diagnosis?

Answer

Screening is a brief examination of an apparently well population to find people who probably have a condition and need full assessment. It produces a referral, never a diagnosis, and the client or parent must be told this plainly. Screening helps only if the referral leads to accessible care.

#296 · Prevention Education and Health Promotion

How is a screening tool judged, and how are school sealant programs targeted?

Answer

By sensitivity, the ability to detect those who have the disease, and specificity, the ability to exclude those who do not. School sealant programs cover permanent first and second molars, often with portable equipment, and targeting whole schools by neighbourhood disadvantage reaches high risk children more efficiently than screening every child.

#297 · Prevention Education and Health Promotion

What is community water fluoridation and how does it work?

Answer

It adjusts the fluoride already present in a public water supply to a caries reducing concentration, which Health Canada sets at an optimal 0.7 mg/L, with a separate and much higher maximum acceptable concentration for safety. The benefit is mainly topical, promoting remineralization and inhibiting demineralization, with a smaller pre eruptive effect.

#298 · Prevention Education and Health Promotion

What is the equity argument for water fluoridation?

Answer

It reaches everyone connected to the supply regardless of income, insurance, education or dental attendance, and needs no appointment, compliance or purchase. Its benefit is largest in the lowest income groups, so it narrows the social gradient rather than widening it as individual measures tend to do.

#299 · Prevention Education and Health Promotion

What does client level advocacy look like for a dental hygienist?

Answer

Acting for the person in front of you: helping them apply to a public plan, arranging interpretation, adjusting an appointment to fit a shift or a transit route, writing a letter supporting a predetermination, or challenging a facility that is not providing daily oral care to a resident.

#300 · Prevention Education and Health Promotion

What is policy level advocacy, and what boundaries apply to advocacy?

Answer

Acting on the conditions that produced the problem, such as presenting local screening data to a council debating fluoridation or submitting to consultations on long term care standards. Advocacy stays within professional standards for accurate representation, respects the client's own priorities, is evidence informed, and partners with communities rather than speaking for them.

#301 · Clinical Therapy

What are the six components of the dental hygiene process of care?

Answer

Assessment, dental hygiene diagnosis, planning, implementation, evaluation and documentation. Documentation is not a final step, it runs through the other five, because a finding never recorded cannot support a diagnosis and care never recorded cannot be defended. The cycle is continuous, not linear.

#302 · Clinical Therapy

How is the recall interval chosen for a client?

Answer

By documented risk, not by habit. Low risk clients with stable tissue are typically seen at 9 to 12 months, moderate risk clients at 6 months, and high risk clients including treated periodontitis, smokers and those with poorly controlled diabetes at 3 to 4 months.

#303 · Clinical Therapy

What wording pattern does a dental hygiene diagnosis follow?

Answer

Problem, related to cause, as evidenced by signs and symptoms. For example, localized dentin hypersensitivity at 13, 14 and 15 related to toothbrush abrasion and 3 mm of buccal recession, as evidenced by a sharp transient response to air challenge.

#304 · Clinical Therapy

How does a dental hygiene diagnosis differ from a dental diagnosis?

Answer

A dental diagnosis names a disease requiring treatment within dentistry and only a dentist makes it, such as caries in the distal of 26. A dental hygiene diagnosis names an actual or potential problem the hygienist may treat, prevent or refer, describing the client's response to a condition.

#305 · Clinical Therapy

Why is a statement that the client needs scaling not a hygiene diagnosis?

Answer

It names an intervention, not a problem. A hygiene diagnosis states a problem within hygiene scope, its related cause, and the evidence, such as generalized plaque-induced gingivitis related to inadequate interdental biofilm removal, as evidenced by bleeding on probing at 46 percent of sites.

#306 · Clinical Therapy

When is the health history taken and what does it cover?

Answer

It is taken before any instrument enters the mouth and is reviewed and re-signed at every recall. It covers medical conditions and surgeries, all medications including natural health products, allergies with the exact reaction, pregnancy status, and tobacco, alcohol and substance use.

#307 · Clinical Therapy

Which oral effects of common medications alter dental hygiene care?

Answer

Xerostomia from anticholinergics, antidepressants, antihypertensives and diuretics; gingival enlargement from phenytoin, cyclosporine and nifedipine; increased bleeding from warfarin and antiplatelet agents; and candidiasis from inhaled corticosteroids. Medications matter twice, for the condition they reveal and for their oral effects.

#308 · Clinical Therapy

Why must an allergy be recorded together with its exact reaction?

Answer

Because the reaction sets the risk category. A rash after penicillin and throat swelling after penicillin sit in different levels of danger and lead to different precautions and different drug choices. Recording only the drug name loses the information needed to plan safe care.

#309 · Clinical Therapy

When are blood pressure and pulse taken?

Answer

At the first appointment, at recall, before local anesthetic or any invasive procedure, whenever the health history changes, and whenever the client reports symptoms. A baseline reading allows any later change to be interpreted rather than guessed at.

#310 · Clinical Therapy

How are chairside blood pressure readings below 180/110 mmHg managed?

Answer

Below 130/80 is normal and care proceeds. At 130 to 139 systolic or 80 to 89 diastolic, care proceeds with advice to seek medical follow up. From 140/90 to 179/109 care usually proceeds after the reading is repeated following rest, with stress reduction and a medical referral.

#311 · Clinical Therapy

What is done at a blood pressure reading of 180/110 mmHg or higher?

Answer

Elective care is deferred and the client is referred for prompt medical assessment. If chest pain, severe headache, visual disturbance or neurological signs are also present, this is a hypertensive emergency and emergency medical services are activated immediately.

#312 · Clinical Therapy

What do ASA physical status classes I, II and III describe?

Answer

ASA I is a healthy client. ASA II is mild systemic disease without functional limitation, such as well controlled hypertension, type 2 diabetes, smoking or pregnancy. ASA III is severe systemic disease with substantive functional limitation, such as poorly controlled diabetes or stable angina.

#313 · Clinical Therapy

What are ASA classes IV to VI, and how does ASA status guide treatment?

Answer

ASA IV is a constant threat to life such as unstable angina, ASA V is moribund and ASA VI a brain-dead organ donor. ASA I and II proceed normally, ASA III calls for stress reduction and medical consultation, and ASA IV means elective care is deferred.

#314 · Clinical Therapy

What sequence does the extraoral examination follow?

Answer

The same order every time so nothing is skipped: general appearance and skin colour, head and scalp, facial symmetry, eyes, ears, nose, lips, the temporomandibular joints, muscles of mastication, lymph nodes, thyroid and salivary glands.

#315 · Clinical Therapy

Which lymph node findings suggest infection and which raise concern for malignancy?

Answer

Healthy nodes are not normally palpable. A node that is enlarged, soft, tender, mobile and bilateral suggests a reaction to infection. A node that is enlarged, hard, fixed, non-tender, unilateral and persistent beyond two weeks raises concern for malignancy and needs referral.

#316 · Clinical Therapy

What is assessed and recorded at the temporomandibular joint?

Answer

Palpate the lateral pole of each condyle just anterior to the tragus through opening and closing. Record clicking, popping or crepitus, muscle tenderness, and whether the mandible deviates and returns to the midline or deflects and stays off it. Adult maximum opening is about 40 to 55 mm.

#317 · Clinical Therapy

What sequence does the intraoral soft tissue examination follow?

Answer

Labial mucosa and vestibules, buccal mucosa with the Stensen duct opening opposite 17 and 27, gingiva, dorsal tongue, lateral borders with the tongue drawn forward on gauze, ventral tongue, floor of mouth palpated bimanually, hard palate, then soft palate, uvula, tonsillar pillars and oropharynx.

#318 · Clinical Therapy

How is an oral lesion described properly in the record?

Answer

By exact location and side, size in millimetres, colour, surface such as smooth, granular, papillary, ulcerated or keratotic, consistency such as soft, firm, rubbery or indurated, border as well defined or diffuse, and attachment as sessile or pedunculated. Note whether it is single or multiple and its duration.

#319 · Clinical Therapy

Which oral sites carry the highest risk of squamous cell carcinoma?

Answer

The floor of the mouth, the ventral and lateral tongue, and the soft palate complex including the anterior tonsillar pillar and retromolar area. The lateral tongue and floor of mouth need deliberate retraction, since they are easily under-examined during a rushed screening.

#320 · Clinical Therapy

What are the main risk factors for oral and oropharyngeal cancer?

Answer

Tobacco in any form, alcohol, and the two combined, which is more than additive. High risk human papillomavirus, particularly HPV 16, drives oropharyngeal cancer of the tonsil and base of tongue, often in younger clients with no tobacco history.

#321 · Clinical Therapy

What is the two week rule for a suspicious oral lesion?

Answer

Identify any obvious local cause, remove it, and re-evaluate. Any lesion persisting beyond two weeks after the suspected cause is removed, and any lesion with no identifiable cause, is referred for definitive diagnosis and biopsy. Adjunctive screening devices do not replace biopsy.

#322 · Clinical Therapy

Why is a sharp explorer not forced into questionable pits and fissures?

Answer

A sharp tip can fracture the intact but demineralized surface of a non-cavitated lesion and turn it into a cavity, and can carry cariogenic bacteria between sites, while adding nothing to accuracy. Detection is primarily visual on a clean, dry, well lit tooth, supported by radiographs.

#323 · Clinical Therapy

What do ICDAS codes 0 to 6 grade?

Answer

Coronal lesions visually: 0 sound, 1 an enamel change seen only after air drying, 2 a distinct enamel change seen while wet, 3 localized enamel breakdown, 4 a dark shadow from dentin, 5 a cavity with visible dentin, 6 an extensive cavity. Codes 1 and 2 are managed non-surgically.

#324 · Clinical Therapy

What causes attrition and what does it look like?

Answer

Attrition is tooth surface loss from tooth against tooth contact. It produces flat, shiny wear facets on occlusal and incisal surfaces that match exactly against the opposing tooth, for example matching facets on 16 and 46. Bruxism accelerates it.

#325 · Clinical Therapy

What causes abrasion and what is its classic pattern?

Answer

Abrasion is loss from friction with a foreign object. The classic pattern is a horizontal V-shaped cervical notch, wider than it is deep, from a hard toothbrush, an abrasive dentifrice and horizontal scrubbing, often heaviest on canines and premolars such as 13 and 14.

#326 · Clinical Therapy

What causes erosion and which surfaces does it affect?

Answer

Erosion is chemical dissolution by acid of non-bacterial origin. Intrinsic acid from reflux, vomiting or an eating disorder typically affects the palatal surfaces of 11, 12, 21 and 22. Extrinsic acid from citrus, soft drinks or wine affects facial and occlusal surfaces. Eroded surfaces look smooth, glossy and cupped.

#327 · Clinical Therapy

What is abfraction and why is the mechanism debated?

Answer

Abfraction is a wedge-shaped cervical defect with a sharp internal line angle, deep and narrow, attributed to occlusal loading flexing the cervical enamel. The theory is contested, and most cervical lesions such as those seen at 14 and 15 are now considered multifactorial.

#328 · Clinical Therapy

How is dentin hypersensitivity defined and explained?

Answer

A short sharp pain from exposed dentin on thermal, evaporative, tactile, osmotic or chemical stimuli, once caries, a cracked tooth and pulpitis are excluded. The hydrodynamic theory explains it: fluid movement in patent tubules distorts the odontoblast layer and stimulates A-delta fibres. Lesion localization and lesion initiation must both be present.

#329 · Clinical Therapy

How is tooth mobility tested and graded?

Answer

Tested with two rigid instrument handles on opposite sides of the crown, never a fingertip, which compresses tissue. Grade 0 is physiologic movement only, Grade I up to about 1 mm horizontally, Grade II more than 1 mm horizontally, and Grade III adds vertical depressibility in the socket.

#330 · Clinical Therapy

How does fremitus differ from mobility?

Answer

Fremitus is vibration felt with a fingertip on the facial surfaces of the maxillary teeth while the client taps and grinds, and it indicates occlusal trauma. Class I is barely palpable, Class II palpable but not visible, Class III visible. Mobility is measured with the teeth apart, fremitus in function.

#331 · Clinical Therapy

How is the Angle molar relationship classified?

Answer

By the mesiobuccal cusp of the maxillary first molar against the buccal groove of the mandibular first molar. In Class I the mesiobuccal cusp of 16 occludes with the buccal groove of 46. In Class II the mandibular molar sits distal to that position, in Class III mesial to it.

#332 · Clinical Therapy

Why is Angle Class II not the same thing as a deep overbite?

Answer

Angle classification describes the anteroposterior relationship of the permanent first molars and canines, not incisor overlap. A deep overbite is not Class II and an anterior crossbite is not Class III. A client can have a Class I molar relationship with a 7 mm overbite.

#333 · Clinical Therapy

How is the canine relationship classified and what are the Class II divisions?

Answer

In Class I the maxillary canine occludes in the embrasure between the mandibular canine and first premolar, for example 13 against the 43 and 44 embrasure; Class II is mesial to it, Class III distal. Class II Division 1 has proclined maxillary incisors and increased overjet, Division 2 retroclined maxillary central incisors.

#334 · Clinical Therapy

What are normal overjet and overbite values, and how is each measured?

Answer

Overjet is the horizontal distance from the labial surface of the mandibular incisor to the incisal edge of the maxillary incisor, normally 2 to 3 mm. Overbite is the vertical overlap of the mandibular incisor by the maxillary incisor, normally 1 to 2 mm. Record both separately from molar and canine class.

#335 · Clinical Therapy

What is acquired pellicle and how is it removed?

Answer

A thin, acellular, bacteria-free film of salivary glycoproteins that forms on a cleaned tooth within minutes. It protects against acid and also provides the receptors bacteria attach to. It cannot be rinsed away, is removed only by polishing, and re-forms quickly.

#336 · Clinical Therapy

How are dental biofilm, materia alba and food debris distinguished?

Answer

Dental biofilm is a structured microbial community in an extracellular polymeric matrix, adherent and not removable by rinsing. Materia alba is a soft white unstructured accumulation of bacteria, salivary proteins, epithelial cells and food debris, and it is displaced by a water spray. Food debris is loose material cleared by saliva.

#337 · Clinical Therapy

How do supragingival and subgingival calculus differ?

Answer

Supragingival calculus is white to yellow and chalky, takes its mineral from saliva, and is heaviest on the lingual of 33 to 43 and the buccal of 16 and 26. Subgingival calculus is dark brown to black, dense and flint-like, occurs anywhere subgingivally, and takes its mineral from crevicular fluid.

#338 · Clinical Therapy

How do extrinsic and intrinsic stains differ?

Answer

Extrinsic stain sits on the external surface: brown from tea, coffee and tobacco, black line stain from chromogenic bacteria, green stain on maxillary anterior teeth in children. Intrinsic stain lies within the tooth, such as tetracycline banding, fluorosis or a necrotic pulp darkening a traumatized 11. Only extrinsic stain responds to polishing.

#339 · Clinical Therapy

Under which headings is the gingiva described, and what best signals inflammation?

Answer

Colour, contour, consistency, texture, position and bleeding. Healthy gingiva is pale or coral pink with knife-edged margins, pointed papillae and firm resilient tissue, and melanin pigmentation is a normal variant. Inflamed gingiva is red or bluish-red with rolled margins and bulbous papillae. Bleeding on probing is the most reliable objective sign.

#340 · Clinical Therapy

How is Periodontal Screening and Recording performed and scored?

Answer

The probe has a 0.5 mm ball tip and a colour band from 3.5 to 5.5 mm. The mouth is divided into six sextants and only the highest code per sextant is recorded. Code 3 means the band is partly visible, and code 4 means it disappears entirely. Screening is triage, not full charting.

#341 · Clinical Therapy

What are the four component tissues of the periodontium?

Answer

Gingiva, periodontal ligament, cementum and alveolar bone. Only the gingiva is visible, so assessment relies on probing and radiographs. Cementum is a thin avascular root covering anchoring Sharpey fibres, thinnest at the cementoenamel junction, and loss of the crestal lamina dura is an early radiographic sign of breakdown.

#342 · Clinical Therapy

Describe the gingival zones and the periodontal ligament fibre groups.

Answer

Free gingiva forms the unattached sulcus wall, attached gingiva is bound to bone and cementum, and the mucogingival junction separates it from alveolar mucosa. The ligament fibre groups are alveolar crest, horizontal, oblique, apical and interradicular, with oblique fibres carrying most vertical force.

#343 · Clinical Therapy

List the dentogingival junction from coronal to apical, and the healthy sulcus depth.

Answer

Sulcus, junctional epithelium, then the supracrestal connective tissue attachment. The sulcus is unattached and probes 1 to 3 mm in health. Readings deeper than 3 mm suggest inflammation or attachment loss rather than a normal sulcus.

#344 · Clinical Therapy

How does junctional epithelium attach, and why does inflammation deepen probing readings?

Answer

It attaches by hemidesmosomes and a basal lamina and is the most permeable epithelium in the mouth. In health the probe stops within it. When inflamed it loses integrity and the probe passes into connective tissue, so probing depth overstates the histologic sulcus.

#345 · Clinical Therapy

What is the supracrestal tissue attachment and why does it matter restoratively?

Answer

Junctional epithelium plus the connective tissue attachment beneath it, historically called biologic width, measuring about 2 mm with roughly 1 mm of each. A crown margin on tooth 26 placed into this zone causes persistent inflammation and bone loss.

#346 · Clinical Therapy

What initiates periodontal disease and what does dysbiosis describe?

Answer

Biofilm is the initiating factor: without it there is no plaque induced gingivitis and no periodontitis. Mature subgingival biofilm shifts toward gram negative anaerobes such as Porphyromonas gingivalis, Tannerella forsythia and Treponema denticola. Dysbiosis describes a shift in community balance, not one pathogen arriving.

#347 · Clinical Therapy

Which host mediators cause tissue breakdown, and what controls bone resorption?

Answer

Neutrophils release reactive oxygen species and enzymes, macrophages and lymphocytes release interleukin 1 beta, interleukin 6, tumour necrosis factor alpha and prostaglandin E2, and matrix metalloproteinases degrade collagen. Bone resorption reflects the RANKL to osteoprotegerin ratio, not bacterial load.

#348 · Clinical Therapy

Why can a smoker show low bleeding scores while disease progresses?

Answer

Nicotine causes vasoconstriction and fibrosis, so bleeding on probing is suppressed and the gingiva can look tight and pale while destruction continues. Smoking is the strongest modifiable risk factor, and it also impairs neutrophil function and fibroblast attachment, so healing after debridement is poorer.

#349 · Clinical Therapy

How is diabetes related to periodontitis, and what other risk factors matter?

Answer

The relationship is bidirectional: poor glycemic control impairs healing while periodontal inflammation raises insulin resistance, so control rather than diagnosis modifies risk. Genetics, stress, obesity and xerostomic drugs add risk, and phenytoin, cyclosporine and nifedipine cause drug influenced gingival enlargement.

#350 · Clinical Therapy

Why are chronic and aggressive periodontitis no longer used as diagnoses?

Answer

The 2017 World Workshop retired them because no reliable biological distinction between the two could be shown. Its classification is the current standard and NDHCE items use its language: one periodontitis entity described by stage, extent and grade rather than by those older labels.

#351 · Clinical Therapy

Define periodontal health under the 2017 classification.

Answer

On an intact periodontium, health means no attachment loss, probing depths of 3 mm or less, and bleeding at fewer than 10 percent of sites. Health also exists on a reduced periodontium, including a stable treated client, who remains a periodontitis client for life.

#352 · Clinical Therapy

How is gingivitis defined and when is it called generalized?

Answer

Gingivitis is reversible inflammation confined to the gingiva, with no attachment loss beyond pre-existing loss. It is localized when 10 to 30 percent of sites bleed and generalized above 30 percent. Non biofilm induced gingival diseases, such as lichen planus, do not resolve with biofilm control alone.

#353 · Clinical Therapy

What findings are required to call a case periodontitis?

Answer

Interdental clinical attachment loss at two or more non adjacent teeth, or buccal or lingual attachment loss of 3 mm or more with probing depths over 3 mm at two or more teeth. Every case then carries a stage, an extent and a grade.

#354 · Clinical Therapy

What determines periodontitis stage, and what are the extent descriptors?

Answer

Stage severity comes from interdental clinical attachment loss at the worst site, radiographic bone loss, and tooth loss due to periodontitis. Extent is recorded as localized, generalized or molar incisor pattern. Stage is what you found; grade is how fast it got there.

#355 · Clinical Therapy

Give the thresholds that define Stage I and Stage II periodontitis.

Answer

Stage I is 1 to 2 mm interdental clinical attachment loss with coronal third bone loss under 15 percent. Stage II is 3 to 4 mm attachment loss with coronal third bone loss of 15 to 33 percent. Neither involves tooth loss from periodontitis.

#356 · Clinical Therapy

What separates Stage III from Stage IV periodontitis?

Answer

Both show 5 mm or more interdental clinical attachment loss with bone loss reaching the middle third of the root or beyond. Stage III has four or fewer teeth lost to periodontitis, while Stage IV has five or more teeth lost.

#357 · Clinical Therapy

How do complexity factors change the assigned stage?

Answer

They raise the stage but never lower it. Probing depths of 6 mm or more, vertical bone loss of 3 mm or more, Class II or III furcation and moderate ridge defects give Stage III. Masticatory dysfunction, mobility grade 2 or more, severe ridge defects or fewer than 20 remaining teeth give Stage IV.

#358 · Clinical Therapy

How is periodontitis grade determined?

Answer

Grade reflects the rate of progression. Direct evidence is bone or attachment loss over five years. Otherwise divide percentage radiographic bone loss at the worst site by age: under 0.25 is Grade A, 0.25 to 1.0 is Grade B, over 1.0 is Grade C. Destruction exceeding the biofilm present also indicates Grade C.

#359 · Clinical Therapy

How do smoking and diabetes act as grade modifiers?

Answer

They only raise the grade, never lower it, so a client who neither smokes nor has diabetes keeps whatever grade the bone loss to age ratio gave. In a current smoker, fewer than 10 cigarettes daily raises the case to at least Grade B and 10 or more daily raises it to Grade C. In a client with diagnosed diabetes, glycated hemoglobin under 7.0 percent raises it to at least Grade B and 7.0 percent or more raises it to Grade C.

#360 · Clinical Therapy

Distinguish peri-implant mucositis from peri-implantitis.

Answer

Peri-implant mucositis is bleeding with inflammation but no progressive bone loss, and it is reversible with biofilm control. Peri-implantitis adds bleeding or suppuration, deeper probing compared with baseline, and radiographic bone loss beyond the initial remodelling seen in peri-implant health.

#361 · Clinical Therapy

How do necrotizing periodontal diseases present and how are they managed?

Answer

Rapid onset pain, punched out ulcerated papillae, a grey pseudomembrane and a fetid odour, clustering with stress, smoking, malnutrition and immunosuppression including HIV. Management is gentle debridement, chlorhexidine rinses and pain control, with metronidazole added only when systemic signs are present.

#362 · Clinical Therapy

Define probing depth and recession, and say why attachment loss matters more.

Answer

Probing depth runs from the gingival margin to the pocket base. Recession is the distance from the cementoenamel junction to a gingival margin apical to it. Clinical attachment loss runs from the cementoenamel junction to the pocket base and is the true measure of destruction, because probing depth alone shifts with swelling and recession.

#363 · Clinical Therapy

How is clinical attachment loss calculated in each of the two cases?

Answer

When the gingival margin is apical to the cementoenamel junction, clinical attachment loss equals probing depth plus recession. When the margin is coronal to the cementoenamel junction, it equals probing depth minus the distance from the cementoenamel junction to the margin. When the margin sits at the junction, it equals probing depth.

#364 · Clinical Therapy

How useful is bleeding on probing for predicting future attachment loss?

Answer

Its value is asymmetric. A single bleeding site poorly predicts future attachment loss, with a positive predictive value near 30 percent, while repeated absence of bleeding predicts stability at close to 98 percent. Smoking suppresses bleeding, so a low score in a smoker means little.

#365 · Clinical Therapy

How are furcation involvement and tooth mobility classified?

Answer

Furcation is probed horizontally with a curved Nabers probe: Class I incipient loss in the flute, Class II partial penetration, Class III through and through under intact gingiva, Class IV through and visible. Mobility, tested with two rigid handles, is Grade 1 to 1 mm, Grade 2 over 1 mm, Grade 3 vertical depressibility.

#366 · Clinical Therapy

State the aims and the endpoints of non surgical periodontal therapy.

Answer

Aims are to remove biofilm and calculus, resolve inflammation, reduce probing depths, gain attachment and create maintainable conditions. Endpoints are probing depths of 4 mm or less, no bleeding on probing, no suppuration, stable or improved attachment, and sustainable self care.

#367 · Clinical Therapy

Why is root planing to a glassy hard surface no longer taught?

Answer

Endotoxin is superficially adherent and removable with light instrumentation, not bound deep within cementum as once believed. Aggressive planing needlessly removes tooth structure and causes hypersensitivity. Root debridement removes biofilm and calculus while conserving tooth structure, so the endpoint is a clean root, not a maximally smoothed one.

#368 · Clinical Therapy

Compare the cross sections and face angles of sickle, universal curette and Gracey.

Answer

A sickle has a triangular cross section with a pointed back, so it stays supragingival. A universal curette is semicircular with a rounded back and toe, its face meeting the lower shank at 90 degrees so both edges work. A Gracey is also semicircular but its face is offset at 60 to 70 degrees.

#369 · Clinical Therapy

Which Gracey cutting edge is used, and how is a Gracey adapted?

Answer

Only the lower or outer cutting edge is used, the longer and more curved of the two, because the face is offset rather than perpendicular to the lower shank. Adapt by making the lower shank parallel to the surface being instrumented.

#370 · Clinical Therapy

Describe the grasp and the fulcrum options for hand instrumentation.

Answer

Use a modified pen grasp, thumb and index finger pads opposite on the handle and the middle finger on the shank to sense vibration. Too tight a grasp blunts tactile sensitivity. Fulcrums are intraoral conventional, cross arch, opposite arch, finger on finger, and extraoral palm up or palm down for maxillary molars.

#371 · Clinical Therapy

What is correct adaptation, and what is the working angulation for calculus removal?

Answer

Adaptation keeps the lower third of the cutting edge, nearest the toe, against the tooth so the heel does not dig into tissue. Insert with the face nearly closed at 0 to 10 degrees, then open to a working angulation of 70 to 80 degrees for calculus removal.

#372 · Clinical Therapy

What happens when blade angulation falls below 45 degrees or exceeds 90 degrees?

Answer

Below 45 degrees the blade burnishes the deposit into a smooth layer instead of removing it, making it harder to detect. Above 90 degrees the cutting edge turns away from the tooth and gouges soft tissue. Calculus removal stays at 70 to 80 degrees.

#373 · Clinical Therapy

How do lateral pressure and stroke direction change during debridement?

Answer

Lateral pressure is very light for assessment strokes, firm for calculus removal, then light again for final debridement. Activation comes from wrist and forearm rotation, not finger flexing. Strokes are short, overlapping pull strokes directed away from the pocket base, and may be vertical, oblique or horizontal.

#374 · Clinical Therapy

When and how is a curette sharpened, and why does dullness matter?

Answer

Sharpen at the first sign of dullness, because a dull blade burnishes deposits and forces heavier pressure. Preserve the internal 70 to 80 degree angle between face and lateral surface: hold a moving flat stone at 100 to 110 degrees to the face, finish on a down stroke, and round the toe.

#375 · Clinical Therapy

Compare magnetostrictive and piezoelectric ultrasonic scalers.

Answer

Magnetostrictive units run at about 18,000 to 45,000 cycles per second with an elliptical tip orbit, so all tip surfaces are active and more heat forms. Piezoelectric units run at about 25,000 to 50,000 cycles per second with linear motion, so the lateral surfaces are most active and less heat forms.

#376 · Clinical Therapy

How is an ultrasonic tip adapted, and when is powered scaling avoided?

Answer

Use the lateral surface at 0 to 15 degrees to the tooth, never point the end into the root, and keep the tip moving, since a stationary tip etches the root. Avoid metal tips on implants, porcelain, composite and glass ionomer, and avoid powered scaling with a communicable respiratory infection.

#377 · Clinical Therapy

Compare sodium bicarbonate, glycine and erythritol air polishing powders.

Answer

Sodium bicarbonate, roughly 40 to 65 micrometres, is moderately abrasive and for supragingival enamel stain only, and its sodium load rules it out in sodium restricted diets, hypertension and renal disease. Glycine at about 25 micrometres and erythritol at about 14 micrometres are low abrasive and safe subgingivally, on roots and implants.

#378 · Clinical Therapy

When is reassessment done after non surgical therapy, and why that window?

Answer

Reassess four to six weeks after completing therapy. Earlier than four weeks the connective tissue has not matured and the result looks worse than it is. Later than six weeks biofilm has recolonized and you are measuring something else. A good response shows bleeding at fewer than 10 percent of sites.

#379 · Clinical Therapy

When should a client be referred to a periodontist?

Answer

When pockets of 6 mm or more persist with bleeding after thorough therapy, and for Stage III or IV disease, Grade C progression, Class II or III furcation, intrabony defects suited to regeneration, mucogingival defects, peri-implantitis, or progression despite adherence. Refer promptly for a young client with rapid destruction.

#380 · Clinical Therapy

What is supportive periodontal therapy and how is its interval set?

Answer

It follows every completed case and repeats medical history, charting, risk reassessment, debridement and self care coaching. Three months is the default, because subgingival biofilm returns toward baseline in roughly nine to eleven weeks. Intervals run three to six months, set by risk, and lengthened only when risk improves.

#381 · Clinical Therapy

How are x-rays produced in a dental tube head?

Answer

Current heats the cathode filament and boils off electrons, called thermionic emission, and high voltage drives them at a tungsten anode target. About one percent of the energy becomes photons, mostly bremsstrahlung, and the rest becomes heat. An aluminium filter removes weak photons and a lead collimator restricts beam size and shape.

#382 · Clinical Therapy

What does kilovoltage peak control, and how does the setting change contrast?

Answer

Kilovoltage controls penetration and contrast. A high setting such as 90 kVp gives low contrast with many greys, a long scale, which suits periodontal bone assessment. A low setting such as 65 kVp gives high contrast, mostly black and white, a short scale, which suits caries detection.

#383 · Clinical Therapy

What do milliamperage and exposure time control?

Answer

Milliamperage sets how many photons are produced per second and time sets how long production continues. Together they give milliampere seconds, the quantity of radiation delivered, and quantity controls density, the overall darkness of the image. Kilovoltage is contrast and penetration; milliamperage and time are quantity and density. Swapping them is a classic trap.

#384 · Clinical Therapy

State the inverse square law and work through a dental example.

Answer

New intensity equals old intensity multiplied by the square of the ratio of old distance to new distance. A beam of 100 units at 20 cm becomes 100 multiplied by 0.25, which is 25 units, at 40 cm. Doubling the distance therefore needs four times the exposure time.

#385 · Clinical Therapy

Distinguish the direct and indirect effects of ionizing radiation.

Answer

In the direct effect a photon strikes a critical molecule, usually DNA, and breaks it. In the indirect effect a photon splits a water molecule into reactive free radicals that then attack nearby molecules. Because the body is mostly water, the indirect effect causes roughly two thirds of the damage from a dental beam.

#386 · Clinical Therapy

What are deterministic radiation effects, and do dental doses reach them?

Answer

Deterministic effects have a dose threshold below which they do not occur, and above it severity rises with dose. Examples are skin erythema, cataract and hair loss. Dental doses fall far below any known threshold, so deterministic effects are not an expected outcome of dental radiography.

#387 · Clinical Therapy

What are stochastic radiation effects, and why do they justify ALARA?

Answer

Stochastic effects have no known threshold, and dose changes the probability of the effect rather than its severity. Cancer induction and heritable effects are stochastic, so no dose is automatically safe and every exposure must be justified. Damage is cumulative over a lifetime because cellular repair is imperfect.

#388 · Clinical Therapy

Which cells and tissues are most radiosensitive, and which matter in dentistry?

Answer

Rapidly dividing undifferentiated cells are most radiosensitive: bone marrow, reproductive cells, the lens of the eye and intestinal lining. In dental imaging the concerns are the thyroid gland, which is sensitive and close to the field, and the parotid and submandibular salivary glands, which lie in the path of many projections.

#389 · Clinical Therapy

How do typical dental doses compare with natural background radiation in Canada?

Answer

A four image digital bitewing series with rectangular collimation delivers roughly 3 to 10 microsieverts, and a panoramic image roughly 10 to 25. Canadian background radiation averages roughly 1800 to 3000 microsieverts per year, about 5 to 8 per day, so a bitewing series is comparable to roughly one day of background.

#390 · Clinical Therapy

What does ALARA stand for and mean?

Answer

ALARA means as low as reasonably achievable, allowing for economic and social factors. It requires that every dental exposure be kept to the smallest dose that still answers the clinical question, rather than simply to a dose below a legal limit. Health Canada's Safety Code 30 applies this principle to dentistry.

#391 · Clinical Therapy

Define justification and optimization in radiation protection.

Answer

Justification means the exposure must do more good than harm for this individual client, decided before any image is taken. Optimization means delivering the lowest dose that still answers the clinical question, through collimation, fast receptors and correct technique. Both principles underpin Health Canada's Safety Code 30 and provincial radiation requirements.

#392 · Clinical Therapy

Which technical choices most reduce client dose?

Answer

Rectangular collimation shapes the beam to receptor size and roughly halves the exposed tissue area. Fast receptors, meaning digital sensors, phosphor plates or F speed film rather than D speed, cut exposure further. Correct alignment prevents retakes, and every retake doubles the dose for that projection.

#393 · Clinical Therapy

What is the current position on lead aprons and thyroid collars?

Answer

Recent guidance concludes that with rectangular collimation, fast receptors and correct alignment they add very little, because dose to shielded organs comes almost entirely from internal scatter. Canadian practice is in transition and provincial regulations still direct their use, so follow the provincial requirement and respect a client who requests a collar.

#394 · Clinical Therapy

Where should the operator stand during an exposure, and how is dose monitored?

Answer

Stand at least two metres from the tube head at 90 to 135 degrees to the primary beam, the zone of least scatter, or behind a fixed barrier, and never in the primary beam. Personal dosimetry monitors occupational dose and is required for dental x-ray workers in most jurisdictions.

#395 · Clinical Therapy

May an operator hold a receptor or steady the tube head during exposure?

Answer

No. No operator ever holds a receptor in a client's mouth or steadies the tube head during an exposure, because that places the operator in the primary beam. If a client cannot hold the receptor, a receptor holding device is used, and if needed a caregiver rather than staff assists.

#396 · Clinical Therapy

What do selection criteria mean for prescribing radiographs?

Answer

Radiographs are prescribed for an individual after a health history and clinical examination, never on a fixed calendar. Published selection criteria give suggested intervals by age and risk category, and those are guidance to be applied to the case in front of you, not a schedule to be followed automatically.

#397 · Clinical Therapy

Why is routine time based prescribing of bitewings not defensible?

Answer

Taking bitewings for every client every six or twelve months regardless of findings exposes low risk clients to radiation that will not change their care, and in most provinces it is a standards violation. A competent client may refuse after being told what the images would show and the consequence of declining.

#398 · Clinical Therapy

Describe the paralleling technique and why it is preferred.

Answer

The receptor is placed parallel to the long axis of the tooth with the central ray perpendicular to both, so length is reproduced accurately. The receptor sits away from the tooth, so a long position indicating device of 30 cm or more offsets magnification. It gives the truest alveolar crest, which periodontal assessment needs.

#399 · Clinical Therapy

Describe the bisecting angle technique and its indications and limits.

Answer

The receptor rests against the tooth and the central ray is directed perpendicular to an imaginary line bisecting the angle between receptor and tooth. It suits a shallow palate, a strong gag reflex, tori and young children. It relies on an estimated line, distorts more readily, and is unsuitable for accurate bone assessment.

#400 · Clinical Therapy

How is a standard bitewing taken, and what does each projection capture?

Answer

The receptor sits lingual to the teeth and the central ray passes through the contacts at about plus 10 degrees vertical. A premolar bitewing captures from the distal of 13 or 43 back through the first molar; a molar bitewing is centred on the second molar and captures the distal of the second premolar with 16, 17 and 18.

#401 · Clinical Therapy

When are vertical bitewings preferred over horizontal bitewings?

Answer

Vertical bitewings turn the receptor so its long axis is vertical, capturing more bone height. They are preferred for moderate to advanced periodontal bone loss, because a horizontal bitewing can cut off the alveolar crest and leave the true extent of bone loss unrecorded.

#402 · Clinical Therapy

What causes elongation on a periapical image, and how is it corrected?

Answer

Elongation means the image is longer than the actual tooth. It is caused by insufficient vertical angulation, and the correction is to increase the vertical angulation before retaking. The memory aid is that an insufficient angle gives a long image.

#403 · Clinical Therapy

What causes foreshortening on a periapical image, and how is it corrected?

Answer

Foreshortening means the image is shorter than the actual tooth. It is caused by excessive vertical angulation, and the correction is to decrease the vertical angulation before retaking. The memory aid is that an excessive angle gives a short image.

#404 · Clinical Therapy

What is a cone cut and what causes it?

Answer

A cone cut is a clear unexposed area on the image where the beam missed part of the receptor. It is caused by misalignment of the position indicating device with the receptor. The correction is to centre the beam over the receptor using an aiming ring or external aiming device.

#405 · Clinical Therapy

What causes overlapping proximal contacts, and why does it matter?

Answer

Overlapping is caused by incorrect horizontal angulation, and the correction is to direct the central ray straight through the contact area. It matters because superimposed proximal surfaces make interproximal caries and crestal bone impossible to assess, which defeats the purpose of a bitewing.

#406 · Clinical Therapy

What do a herringbone pattern, a blurred image and a double exposure indicate?

Answer

A light image with a herringbone pattern means the receptor was reversed and the beam entered through the lead foil backing; place the active surface toward the beam and retake. Blurring comes from movement of client, tube head or receptor. A double exposure means one receptor was exposed twice, so separate exposed from unexposed receptors.

#407 · Clinical Therapy

How are density errors recognized and corrected?

Answer

Density is overall darkness and density errors are exposure errors. An image that is too dark received too much exposure, so reduce time or milliamperage. An image that is too light received too little exposure, so increase time or milliamperage. Density is not corrected by changing kilovoltage.

#408 · Clinical Therapy

How are contrast errors recognized and corrected?

Answer

Contrast errors are kilovoltage errors. A flat grey image with too many similar shades had kilovoltage set too high, so lower it. A harshly black and white image that loses detail in the shadows had kilovoltage set too low, so raise it.

#409 · Clinical Therapy

Compare direct and indirect digital imaging systems.

Answer

Direct systems use a rigid solid state sensor, either charge coupled device or CMOS, giving an image within seconds, so holders are essential. Indirect systems use a photostimulable phosphor plate read by a laser scanner. Plates are thin, flexible and film sized, but need a scanning step and are erased by light.

#410 · Clinical Therapy

What image enhancement is acceptable, and what counts as fraud?

Answer

Reasonable adjustment of brightness, contrast and magnification is acceptable, the original file is retained, and the record notes the enhancement. Altering an image to misrepresent what it shows, such as painting out a lesion or sending an edited image to a third party payer, is fraud. Poor images are retaken, never edited.

#411 · Clinical Therapy

What is panoramic imaging good for, and what can it not replace?

Answer

A tube head and receptor rotate in opposite directions, imaging both jaws and the temporomandibular joints, with only structures in the focal trough in focus. It assesses third molars 18, 28, 38 and 48, large lesions, impactions and implant planning. Resolution is low, so it never replaces bitewings for interproximal caries.

#412 · Clinical Therapy

How do chin and front to back positioning errors appear on a panoramic image?

Answer

Chin too high flattens the occlusal plane into a frown and superimposes the hard palate over the maxillary apices. Chin too low exaggerates it into a deep smile and blurs the anterior teeth. Standing too far forward makes anterior teeth narrow and blurred, too far back makes them wide and magnified.

#413 · Clinical Therapy

What is a ghost image on a panoramic radiograph?

Answer

A ghost image occurs when a dense object sits between the source and the centre of rotation, so the beam records it twice. The ghost appears on the opposite side of the image from the real object, higher, larger and blurred. Earrings and metal jewellery produce ghosts, so remove all removable metal first.

#414 · Clinical Therapy

What is cone beam computed tomography used for, and what limits its use?

Answer

A cone shaped beam builds a three dimensional volume showing the mandibular canal, the position of an impacted tooth such as 23, and root fractures. Dose is roughly 20 to 100 microsieverts for a small field of view, so it is prescribed only when conventional imaging cannot answer the question, never for screening.

#415 · Clinical Therapy

What do radiolucent and radiopaque mean, with examples?

Answer

Radiolucent means dark, because the structure let the beam through: air, caries, a periapical lesion, the pulp chamber and a foramen. Radiopaque means light, because the structure absorbed radiation: enamel, dentin, cortical bone, calculus and metallic restorations.

#416 · Clinical Therapy

Name the normal landmarks most often mistaken for disease.

Answer

The incisive foramen is an oval radiolucency between the roots of 11 and 21, and the mental foramen is often projected near the apex of 35 or 45, a classic mimic of a periapical lesion. Also expect the maxillary sinus, the zygomatic process over 16 or 26, and the mandibular canal.

#417 · Clinical Therapy

How does interproximal caries appear radiographically, and how is depth graded?

Answer

It appears as a radiolucent notch on the proximal surface just apical to the contact point, classically triangular with its base at the outer enamel surface. A lesion in the outer half of enamel is incipient, one more than halfway through enamel but not reaching the dentinoenamel junction is moderate, one crossing that junction into dentin is advanced, and one more than halfway from the junction to the pulp is severe. Radiographs underestimate true depth.

#418 · Clinical Therapy

How do calculus and defective restorations appear on a radiograph?

Answer

Calculus appears as pointed radiopaque spurs or collars on proximal root surfaces, and radiographs show only a fraction of what is present because facial and lingual deposits are hidden by the tooth. Defective restorations show open or overhanging margins, an overhang appearing as a radiopaque projection beyond the crown contour.

#419 · Clinical Therapy

Distinguish horizontal from vertical bone loss radiographically.

Answer

Normally the crest sits 1 to 2 mm apical to the cementoenamel junctions, parallel to a line joining them. In horizontal loss the crest stays parallel to that line but further apical, affecting adjacent teeth evenly. In vertical or angular loss the crest is not parallel, and a defect angles down one root surface.

#420 · Clinical Therapy

What may a dental hygienist report from a radiograph, and what may they not?

Answer

The hygienist may record what is visible, including bone levels, calculus, apparent caries and defective margins, and may use those findings in the dental hygiene diagnosis and care plan. The hygienist does not give a definitive diagnosis of caries or pathology, and refers a suspicious radiolucency to the dentist promptly, recording the referral.

#421 · Clinical Therapy

What is the difference between pharmacokinetics and pharmacodynamics?

Answer

Pharmacokinetics is what the body does to a drug: absorption, distribution, hepatic metabolism by cytochrome P450 enzymes, and mostly renal excretion. Pharmacodynamics is what the drug does to the body. Route governs speed: intravenous is immediate, inhaled and sublingual are fast because they avoid first pass metabolism, and oral is slowest.

#422 · Clinical Therapy

Define therapeutic index, agonist and antagonist.

Answer

Therapeutic index is the ratio of toxic dose to effective dose, and warfarin has a narrow one. An agonist binds a receptor and produces a response. An antagonist blocks the receptor, as naloxone does at opioid receptors. Half life is the time for plasma concentration to halve.

#423 · Clinical Therapy

How do you tell a drug side effect from a true allergy?

Answer

A side effect is predictable and dose related but not the therapeutic goal, such as xerostomia from an anticholinergic. An adverse effect is harmful and unintended. An allergy is immune mediated and not dose related, so nausea after codeine is a side effect while hives and wheeze is an allergy.

#424 · Clinical Therapy

Which drugs cause gingival enlargement, and what is the primary response?

Answer

Calcium channel blockers, mainly nifedipine but also amlodipine and diltiazem, plus phenytoin and cyclosporine. The enlargement begins at the interdental papillae and worsens where biofilm accumulates, so meticulous biofilm control is the primary response. Ask the prescriber about a medication review rather than telling the client to stop.

#425 · Clinical Therapy

How do you manage a client taking an anticoagulant before scaling?

Answer

Warfarin is monitored by the international normalized ratio, usual therapeutic range 2.0 to 3.0, and scaling and root planing can generally proceed within that range using local hemostatic measures. Apixaban, rivaroxaban and dabigatran are not INR monitored. Never advise a client to stop an anticoagulant or antiplatelet.

#426 · Clinical Therapy

What are antiresorptive drugs, and how is MRONJ defined?

Answer

Antiresorptives are bisphosphonates such as alendronate, risedronate and intravenous zoledronic acid, plus denosumab; antiangiogenics such as bevacizumab behave similarly. MRONJ is exposed bone, or bone probeable through a fistula in the maxillofacial region, persisting beyond eight weeks. Non surgical periodontal therapy is not contraindicated. Refer if exposed bone appears.

#427 · Clinical Therapy

Which drugs cause xerostomia, and how is it managed?

Answer

Tricyclic antidepressants, serotonin reuptake inhibitors, antihistamines, antipsychotics, bladder antimuscarinics and diuretics dry the mouth. Hyposalivation is unstimulated flow below about 0.1 mL per minute. Advise frequent water, sugar free xylitol gum, saliva replacement gels, and 5000 ppm sodium fluoride paste daily with varnish at short recall intervals.

#428 · Clinical Therapy

Why do inhaled corticosteroids cause oral problems, and what advice helps?

Answer

Inhaled corticosteroids suppress local immunity on the oropharyngeal mucosa and predispose the client to candidiasis, seen as white plaques that wipe off, or as angular cheilitis. Advise the client to rinse the mouth and spit after every dose, and to use a spacer where one has been prescribed.

#429 · Clinical Therapy

How do local anesthetics block nerve conduction?

Answer

They bind the intracellular portion of voltage gated sodium channels, so sodium cannot enter, the membrane cannot reach threshold, and the impulse cannot propagate. The uncharged free base is the only form that crosses the lipid nerve sheath, and the ionized cation is the form that binds the receptor inside.

#430 · Clinical Therapy

Why is an inflamed tooth harder to freeze?

Answer

Inflamed tissue is acidic, so more anesthetic exists as the ionized cation and less free base crosses the nerve sheath, making the block slower and weaker. Inflammation also raises blood flow, which carries drug away, and sensitizes nociceptors. A regional block away from the site beats infiltrating into it.

#431 · Clinical Therapy

Compare ester and amide local anesthetics, and where does articaine sit?

Answer

Esters such as benzocaine, procaine and tetracaine are hydrolyzed in plasma to para aminobenzoic acid, a recognized allergen, so true allergy is more common and they survive mainly as topicals. Amides such as lidocaine, mepivacaine, prilocaine and bupivacaine are metabolized hepatically with rare allergy. Articaine is an amide with an ester side chain.

#432 · Clinical Therapy

What does adding a vasoconstrictor to a local anesthetic achieve?

Answer

Epinephrine slows systemic absorption, so peak plasma concentration and toxicity risk fall, the block lasts longer and goes deeper, and bleeding at the site is reduced. Those cartridges also contain sodium metabisulphite as a preservative, which matters for a sulphite sensitive client.

#433 · Clinical Therapy

Name the Canadian local anesthetic agents and their concentrations.

Answer

Lidocaine 2 percent with epinephrine 1:100,000 or 1:50,000, about 60 minutes pulpal. Articaine 4 percent with epinephrine 1:100,000 or 1:200,000, 60 to 75 minutes pulpal. Mepivacaine 3 percent plain, 20 to 40 minutes. Prilocaine 4 percent plain or with 1:200,000. Bupivacaine 0.5 percent with 1:200,000, avoided in children.

#434 · Clinical Therapy

How do you convert a percentage solution to milligrams per cartridge?

Answer

Multiply the percentage by 10 for milligrams per millilitre, because 1 percent is 1000 mg in 100 mL. So 2 percent is 20 mg per mL, 3 percent 30, 4 percent 40, 0.5 percent 5. Multiply by the 1.8 mL Canadian cartridge: 36 mg, 54 mg, 72 mg and 9 mg.

#435 · Clinical Therapy

How much epinephrine is in one 1.8 mL cartridge at each dilution?

Answer

A 1:100,000 dilution is 0.01 mg per mL, giving 0.018 mg per 1.8 mL cartridge. A 1:200,000 dilution is 0.005 mg per mL, giving 0.009 mg per cartridge. A 1:50,000 dilution is 0.02 mg per mL, giving 0.036 mg per cartridge.

#436 · Clinical Therapy

State the adult maximum recommended doses of the common local anesthetics.

Answer

Lidocaine with epinephrine 7.0 mg per kg to a maximum of 500 mg. Articaine 7.0 mg per kg to 500 mg, reduced to 5 mg per kg in children. Mepivacaine 6.6 mg per kg to 400 mg. Prilocaine 8.0 mg per kg to 600 mg. Bupivacaine about 1.3 mg per kg to 90 mg.

#437 · Clinical Therapy

What limits epinephrine per appointment, and which limit governs dosing?

Answer

Epinephrine is limited to 0.2 mg per appointment in a healthy adult and 0.04 mg in significant cardiovascular disease, where 1:50,000 is avoided. Calculate both the drug limit in mg per kg and the epinephrine limit, then use whichever gives the lower number of cartridges.

#438 · Clinical Therapy

How many cartridges for a healthy 70 kg adult on lidocaine 2 percent?

Answer

With 1:100,000 epinephrine the drug limit is 7.0 x 70 = 490 mg, under the 500 mg ceiling, and 490 divided by 36 mg per cartridge is 13.6, so 13 cartridges. The epinephrine limit is 0.2 divided by 0.018, which is 11.1, so 11 cartridges. Eleven is the answer.

#439 · Clinical Therapy

How many articaine cartridges for a 60 kg client with cardiovascular disease?

Answer

Articaine 4 percent with 1:200,000 epinephrine. Drug limit 7.0 x 60 = 420 mg, and 420 divided by 72 mg per cartridge is 5.8, so 5 cartridges. Epinephrine limit 0.04 divided by 0.009 is 4.4, so 4 cartridges, delivering 0.036 mg epinephrine and 288 mg articaine. Four is the answer.

#440 · Clinical Therapy

How does local anesthetic systemic toxicity present, and what is the response?

Answer

Excitatory central nervous system signs come first: circumoral numbness, metallic taste, tinnitus, lightheadedness, slurred speech, twitching, then tonic clonic seizure and respiratory depression. Cardiovascular collapse follows, with bupivacaine the most cardiotoxic agent. Response: stop injecting, supine, maintain the airway, 100 percent oxygen, activate emergency medical services, protect rather than restrain.

#441 · Clinical Therapy

What causes methemoglobinemia, and how is it recognized and treated?

Answer

Prilocaine is the amide most implicated and benzocaine the topical. Hemoglobin iron is oxidized to the ferric state and cannot carry oxygen. Signs appear 1 to 3 hours later: cyanosis unresponsive to oxygen, chocolate brown blood, headache, and oximetry plateauing near 85 percent. Give oxygen, transfer, and hospital intravenous methylene blue.

#442 · Clinical Therapy

Describe paresthesia, hematoma and trismus after local anesthesia.

Answer

Paresthesia is altered sensation persisting after the anesthetic should have worn off, most often lingual after an inferior alveolar block, with 4 percent formulations reported disproportionately; refer beyond about two months. Hematoma follows vessel puncture: pressure at once, ice for a day, 7 to 14 days of discoloration. Trismus needs heat and gentle exercises.

#443 · Clinical Therapy

List the epinephrine cautions before giving a local anesthetic.

Answer

Cardiovascular disease: limit epinephrine to 0.04 mg and avoid 1:50,000. Non selective beta blockers such as propranolol: hypertensive response with reflex bradycardia. Tricyclic antidepressants: limit epinephrine and avoid levonordefrin. Cocaine or methamphetamine within 24 hours: no vasoconstrictor. Uncontrolled hyperthyroidism: avoid epinephrine. In pregnancy do not withhold anesthesia; avoid prilocaine.

#444 · Clinical Therapy

How is nitrous oxide titrated, and what is diffusion hypoxia?

Answer

Start on 100 percent oxygen at 5 to 7 litres per minute, then add nitrous oxide in small increments; most clients are comfortable between 20 and 50 percent. Finish with 100 percent oxygen for at least 3 to 5 minutes, otherwise nitrous oxide dilutes alveolar oxygen and causes headache, nausea and lethargy.

#445 · Clinical Therapy

When is nitrous oxide and oxygen sedation contraindicated?

Answer

Nasal obstruction, since delivery is nasal; severe chronic obstructive pulmonary disease; recent middle ear or intraocular surgery involving a gas bubble; bowel obstruction; vitamin B12 deficiency; recent bleomycin therapy; and the first trimester of pregnancy. Oversedation shows as nausea, sweating, dysphoria and a fixed stare, so reduce the concentration.

#446 · Clinical Therapy

What is first line analgesia for acute dental pain, and at what dose?

Answer

Acute dental pain is inflammatory, so a non steroidal anti inflammatory drug beats an opioid. Ibuprofen 400 mg every 4 to 6 hours is the usual adult dose, ceiling 1200 mg in 24 hours over the counter and 2400 mg on prescription. Ibuprofen 400 mg with acetaminophen 1000 mg beats either alone.

#447 · Clinical Therapy

How is acetaminophen dosed, and when is it preferred over an NSAID?

Answer

Adults take 325 to 1000 mg every 4 to 6 hours, maximum 4000 mg in 24 hours, commonly reduced to 2000 to 3000 mg in older adults, liver disease or regular alcohol use. Children take 10 to 15 mg per kg per dose. Prefer acetaminophen for an anticoagulated client.

#448 · Clinical Therapy

What are the first steps in any dental office medical emergency?

Answer

Stop the procedure, clear the mouth, position the client, assess responsiveness, then airway, breathing and circulation. Send someone for the emergency kit, oxygen and the defibrillator, and note the time. Oxygen is given in every dental emergency except hyperventilation.

#449 · Clinical Therapy

How do you recognize and manage syncope?

Answer

Syncope is transient cerebral hypoperfusion from a vasovagal response. Recognition: pallor, sweating, nausea, dizziness, then brief loss of consciousness with a slow weak pulse. Sequence: stop treatment, supine with the legs slightly elevated, loosen clothing, open the airway, check breathing, oxygen. Activate emergency medical services if consciousness does not return within a minute.

#450 · Clinical Therapy

How is hyperventilation recognized and managed?

Answer

Rapid deep breathing causes respiratory alkalosis, with tingling of the fingers and around the mouth, lightheadedness and carpopedal spasm. Sequence: stop treatment, position the client upright, coach slow breathing at about one breath every 5 seconds, and rebreathe into cupped hands. Withhold oxygen, because the problem is too little carbon dioxide.

#451 · Clinical Therapy

How is an acute asthma attack managed chairside?

Answer

Recognition: wheeze, cough, chest tightness, prolonged expiration, speech in short phrases. Sequence: stop treatment, sit the client upright leaning forward, give their own salbutamol 2 puffs of 100 micrograms with a spacer, oxygen, repeat every few minutes. Activate emergency medical services if speech drops to single words or the chest goes silent.

#452 · Clinical Therapy

How is anaphylaxis recognized, and what is the first line drug and dose?

Answer

Anaphylaxis involves two or more body systems, or hypotension after a known allergen: hives, swelling of the lips, tongue and throat, stridor, wheeze, vomiting, collapse. Epinephrine is first line and is given immediately, 0.3 to 0.5 mg of the 1:1000 solution, 1 mg per mL, intramuscularly into the anterolateral thigh.

#453 · Clinical Therapy

State the anaphylaxis response sequence in the correct order.

Answer

Stop the causative agent. Give epinephrine intramuscularly at once, 0.15 mg for a child under 30 kg. Then activate emergency medical services, position supine with legs elevated, give high flow oxygen, and repeat epinephrine every 5 to 15 minutes without improvement. Epinephrine precedes the phone call, antihistamines and corticosteroids. Everyone treated goes to hospital.

#454 · Clinical Therapy

How do you manage a foreign body airway obstruction?

Answer

If the client can cough forcefully, speak or breathe, encourage coughing and do nothing else. If they cannot speak, cough or breathe, give 5 back blows between the scapulae leaning forward, then 5 abdominal thrusts, alternating until the object clears or they become unresponsive, then start chest compressions. Never sweep blindly.

#455 · Clinical Therapy

How is an angina attack managed in the dental chair?

Answer

Recognition: substernal pressure often radiating to the left arm, jaw or neck. Sequence: stop treatment, position semi upright, oxygen, then their own nitroglycerin 0.3 or 0.4 mg sublingually, repeated every 5 minutes to a maximum of 3 doses. Check first for sildenafil within 24 hours or tadalafil within 48 hours.

#456 · Clinical Therapy

When do you suspect myocardial infarction, and what is the sequence?

Answer

Suspect it when pain is severe, lasts beyond 15 minutes, does not resolve after three nitroglycerin doses, or comes with sweating, nausea or breathlessness; presentation is often atypical in women, older adults and people with diabetes. Activate emergency medical services, oxygen, acetylsalicylic acid 160 to 325 mg chewed, keep the client still.

#457 · Clinical Therapy

How is a suspected stroke recognized and managed?

Answer

FAST: facial droop, arm weakness, speech difficulty, time to call. Stop treatment, activate emergency medical services at once stating stroke is suspected, record the time the client was last known to be well, rest with the head slightly elevated, oxygen, nothing by mouth. Do not give acetylsalicylic acid, since the stroke may be hemorrhagic.

#458 · Clinical Therapy

How is hypoglycemia recognized and treated?

Answer

Hypoglycemia is blood glucose below 4.0 mmol/L, with sweating, pallor, tachycardia, confusion and slurred speech. Conscious: 15 g of fast acting carbohydrate such as 150 mL of juice or 4 glucose tablets, wait 15 minutes, recheck. Unable to swallow: recovery position, oxygen, activate emergency medical services, glucagon 1 mg intramuscularly.

#459 · Clinical Therapy

What is the correct response to a generalized tonic clonic seizure?

Answer

Stop treatment, remove everything from the mouth, lower the chair to supine, clear sharp objects away, protect the head and note the start time. Do not restrain the client and never place anything in the mouth. Afterwards open the airway, check breathing, recovery position, oxygen. Call emergency medical services beyond 5 minutes.

#460 · Clinical Therapy

How is cardiac arrest managed in the dental office?

Answer

Confirm unresponsiveness, activate emergency medical services and send for the defibrillator. Check breathing and a carotid pulse together for no more than 10 seconds, treating gasping as no breathing. With no pulse, compress the lower sternum 5 to 6 cm deep, 100 to 120 per minute, 30 compressions to 2 breaths.

#461 · Clinical Therapy

In FDI notation, which primary teeth erupt first, and when?

Answer

The mandibular central incisors 71 and 81, at 6 to 10 months. Maxillary central incisors 51 and 61 follow at 8 to 12 months, maxillary lateral incisors 52 and 62 at 9 to 13 months, and mandibular lateral incisors 72 and 82 at 10 to 16 months.

#462 · Clinical Therapy

Give the primary eruption sequence and when the dentition is complete.

Answer

Central incisor, lateral incisor, first molar, canine, second molar. First molars 54, 64, 74 and 84 erupt at 13 to 19 months, canines 53, 63, 73 and 83 at 16 to 23 months, second molars 55, 65, 75 and 85 at 23 to 33 months, complete by 30 to 36 months.

#463 · Clinical Therapy

Which permanent teeth erupt first, and when do primary teeth exfoliate?

Answer

The mandibular first molars 36 and 46 and the mandibular central incisors 31 and 41 at 6 to 7 years, along with the maxillary first molars 16 and 26. Primary incisors exfoliate at 6 to 8 years, primary canines and molars between 9 and 12 years.

#464 · Clinical Therapy

How does canine eruption order differ between the two arches?

Answer

Mandibular canines 33 and 43 erupt at 9 to 10 years, before the mandibular premolars 34, 44, 35 and 45. In the maxilla the order reverses: premolars 14, 24, 15 and 25 erupt first, then canines 13 and 23 at 11 to 12 years, which is why maxillary canines are so often left without space.

#465 · Clinical Therapy

When do the permanent incisors, second molars and third molars erupt?

Answer

Maxillary central incisors 11 and 21 and mandibular lateral incisors 32 and 42 at 7 to 8 years, maxillary laterals 12 and 22 at 8 to 9. Second molars 37 and 47 at 11 to 13, 17 and 27 at 12 to 13, third molars 18, 28, 38 and 48 at 17 to 21 years.

#466 · Clinical Therapy

How does primary tooth morphology differ from permanent tooth morphology?

Answer

Primary crowns are whiter, shorter and relatively wider mesiodistally, with marked cervical constriction and a prominent buccal cervical ridge. Enamel and dentin are about half as thick and pulp chambers proportionally larger with high pulp horns, so caries reaches the pulp fast. Contacts are broad and flat, so bitewings are needed.

#467 · Clinical Therapy

Describe tell show do and the other pediatric behaviour guidance techniques.

Answer

Describe the procedure in age-appropriate words, demonstrate it on a fingernail or model, then perform it exactly as shown. Add descriptive praise, distraction, modelling, voice control as a change in tone rather than volume, a stop signal and caregiver presence. Protective stabilization is a last resort needing specific consent.

#468 · Clinical Therapy

Define early childhood caries and describe its typical pattern.

Answer

One or more decayed, missing due to caries, or filled surfaces in any primary tooth in a child under 6 years. The pattern starts on the maxillary primary incisors 51, 52, 61 and 62, then the first molars, sparing the mandibular incisors because the tongue and submandibular ducts protect them.

#469 · Clinical Therapy

List the risk factors for early childhood caries and its management.

Answer

Risk rises with night-time bottle use, sugary drinks in sippy cups, night feeding after eruption, caregiver transmission of cariogenic bacteria and reduced access to care. Management is risk based: a first visit by 12 months, lift-the-lip screening, caregiver brushing, diet counselling, fluoride varnish and silver diamine fluoride.

#470 · Clinical Therapy

State the age-banded fluoride recommendations for young children.

Answer

Under 3 years an adult brushes, using fluoridated toothpaste the size of a grain of rice based on risk assessment. From 3 to 6 years use a green pea and spit without rinsing. Mouthrinse is not recommended under 6 years. Varnish at 5 percent sodium fluoride, about 22 600 ppm, is safest for young children.

#471 · Clinical Therapy

When are pit and fissure sealants indicated, and when are they not?

Answer

Indicated for deep, narrow, retentive pits and fissures, especially newly erupted permanent molars at elevated caries risk, and for noncavitated lesions. Not indicated over a cavitated dentin lesion, over an existing restoration, where the fissure is shallow and self-cleansing, or where the tooth cannot be isolated.

#472 · Clinical Therapy

What is the commonest cause of sealant failure, and why?

Answer

Moisture contamination. Saliva on etched enamel deposits a glycoprotein film within seconds that stops resin flowing into the etched porosities, so the resin tags that provide retention never form. A rubber dam is ideal, and cotton rolls with dry angles and high volume evacuation are acceptable.

#473 · Clinical Therapy

Give the ordered technique for placing a pit and fissure sealant.

Answer

Clean with pumice slurry or air polishing using no oil or fluoride paste, isolate, dry, etch with 37 percent phosphoric acid for 15 to 30 seconds, rinse, re-isolate, dry and confirm a frosty white surface. Apply sealant, draw it through with an explorer, light cure 20 to 40 seconds, check occlusion, then apply varnish.

#474 · Clinical Therapy

What are orthodontic white spot lesions, and how are they managed?

Answer

Demineralized enamel that can appear within four weeks of bonding, usually on the gingival third of the maxillary lateral incisors and canines. Home care needs interdental brushes, floss threaders and a daily 0.05 percent sodium fluoride rinse, with varnish at recall. Remineralization with fluoride and casein phosphopeptide amorphous calcium phosphate comes first.

#475 · Clinical Therapy

What oral signs suggest an eating disorder, and what advice follows?

Answer

Perimylolysis, erosion of the palatal surfaces of teeth 13 to 23, with restorations standing proud of the eroded surface. Look also for painless bilateral parotid enlargement and angular cheilitis. Advise rinsing with water or sodium bicarbonate after an episode rather than brushing, and raise the concern privately.

#476 · Clinical Therapy

Which mouthguard is preferred, and what oral effects does vaping have?

Answer

Custom fabricated mouthguards outperform boil-and-bite and stock guards. Ask adolescents directly about vaping, because the aerosol reduces salivary flow and nicotine vasoconstriction suppresses bleeding on probing, so the tissues look healthy while disease progresses.

#477 · Clinical Therapy

Is dental hygiene care safe during pregnancy?

Answer

Yes, throughout pregnancy, and it should not be deferred. Assessment, scaling, root planing, radiographs, local anesthesia and preventive agents are appropriate in every trimester, since untreated infection is the greater risk. Elective care may be timed for comfort, often the second trimester, but urgent care is never postponed.

#478 · Clinical Therapy

What is supine hypotensive syndrome, and how is it prevented?

Answer

After about 20 weeks a supine position lets the uterus compress the inferior vena cava, producing dizziness, nausea, sweating and falling blood pressure. Prevent it with a semi-supine position, a wedge under the right hip tilting the client to the left, and frequent breaks.

#479 · Clinical Therapy

How are radiographs and local anesthesia handled in pregnancy?

Answer

Radiographs are taken when justified, not withheld, because fetal dose is negligible. The protections that matter are justification, rectangular collimation and fast digital receptors, with a leaded apron and thyroid collar still standard in Canada. Lidocaine 2 percent with epinephrine 1:100 000 is the anesthetic of choice.

#480 · Clinical Therapy

Describe pregnancy gingivitis and the pregnancy granuloma.

Answer

Pregnancy gingivitis is an exaggerated inflammatory response to existing plaque, driven by estrogen and progesterone raising vascular permeability and favouring Prevotella intermedia. It begins around the second month, peaks near the eighth, and resolves after delivery if plaque control holds. A pyogenic granuloma is a painless red mass that bleeds readily and regresses after delivery.

#481 · Clinical Therapy

Which oral changes are true ageing changes and which are not?

Answer

Ageing brings attrition, secondary dentin with a smaller pulp chamber and reduced sensitivity, darker colour, sclerosed tubules and thinner mucosa. Recession, xerostomia and tooth loss are not ageing changes, because they result from disease, medication and trauma, and treating them as inevitable is the error being tested.

#482 · Clinical Therapy

Why does root caries dominate in older adults, and how is it managed?

Answer

Exposed cementum and dentin demineralize at a critical pH of about 6.2 to 6.7 compared with 5.5 for enamel, so root surfaces are lost at pH values that leave enamel untouched. Risk climbs with recession, hyposalivation, partial denture clasps and dexterity loss. Manage with 1.1 percent sodium fluoride, silver diamine fluoride and shorter recalls.

#483 · Clinical Therapy

Distinguish xerostomia from hyposalivation and list the usual causes.

Answer

Xerostomia is the symptom of dry mouth, while hyposalivation is measured reduced flow. Medication is the usual cause: anticholinergics, antidepressants, antihypertensives, diuretics, antihistamines and opioids. Advise frequent water, sugar-free xylitol gum, saliva substitutes and avoidance of alcohol-containing rinses.

#484 · Clinical Therapy

What oral and safety effects does polypharmacy bring in older adults?

Answer

Gingival enlargement from phenytoin, cyclosporine and calcium channel blockers such as nifedipine, plus orthostatic hypotension, so raise the chair slowly. When dexterity fails, adapt the aid rather than repeat the instruction: build up a handle with foam tubing or a tennis ball, add a hand strap, or use a wide-handled power brush.

#485 · Clinical Therapy

How is an appointment adapted for a client living with dementia?

Answer

Book short morning appointments, keep the room quiet, give one instruction at a time and involve a familiar caregiver. Bridging, where the client holds a second brush, and chaining, where the caregiver starts a movement the client finishes, preserve participation. Confirm who holds decision-making authority, since consent may come from a substitute decision maker.

#486 · Clinical Therapy

What is denture stomatitis, and what denture care do you teach?

Answer

Erythema confined to the denture-bearing mucosa, usually Candida albicans with continuous wear and poor denture hygiene. Teach daily denture brushing, overnight removal, storage in water, and no hot water or bleach on a cobalt-chromium framework, since the appliance itself must be disinfected or it recurs. Dysphagia raises aspiration risk, so treat semi-upright with evacuation.

#487 · Clinical Therapy

How do you plan a wheelchair transfer safely?

Answer

Ask the client how they normally transfer. Lock the brakes, swing away the armrest and footrests, position the wheelchair about 30 degrees to the dental chair, lower the dental chair to the same height or slightly lower, and use a transfer belt. Use two people or a sliding board when the client cannot bear weight.

#488 · Clinical Therapy

What is desensitization, and how do you adapt care for autistic clients?

Answer

Desensitization means short repeated non-invasive visits with the same clinician, room and time, adding one step each visit. For autistic clients dim the light, offer sunglasses or noise-reducing headphones, limit ultrasonic noise, avoid strongly flavoured pastes and use visual schedules. Many clients are safest treated in their own wheelchair with headrest support.

#489 · Clinical Therapy

Which diabetes values and appointment timing guide hygiene care?

Answer

Glycated hemoglobin reflects average glycemia over about three months, with a target of 7.0 percent or less for most adults, fasting plasma glucose 4.0 to 7.0 mmol/L and two-hour postprandial 5.0 to 10.0 mmol/L. Book morning appointments after the client has eaten and taken medication, and confirm blood glucose first.

#490 · Clinical Therapy

How do you recognize and treat hypoglycemia in the dental chair?

Answer

Blood glucose below 4.0 mmol/L, with shakiness, sweating, confusion and hunger. Stop treatment, give 15 g of fast-acting carbohydrate and recheck in 15 minutes, repeating if it is still low. A high A1c instead signals impaired healing and infection risk, so consult the physician and shorten the recall interval.

#491 · Clinical Therapy

Is anticoagulation interrupted for routine dental hygiene care?

Answer

No. The thromboembolic risk of stopping the drug exceeds the bleeding risk, which is local and manageable. A warfarin client with a recent INR in therapeutic range, generally up to about 3.5, is scaled without a dose change, and direct oral anticoagulants and antiplatelets continue. Never advise a client to stop an anticoagulant.

#492 · Clinical Therapy

How is care modified in hemophilia or von Willebrand disease?

Answer

Consult the hematologist before treatment and avoid an inferior alveolar nerve block, since a hematoma in the pterygomandibular space can compromise the airway. Use local measures for bleeding, gauze pressure and tranexamic acid mouthrinse, and prefer acetaminophen to NSAIDs for analgesia.

#493 · Clinical Therapy

How is hygiene care timed for an immunocompromised client?

Answer

Timing, not avoidance. Eliminate sources of infection two to three weeks before chemotherapy or transplantation, work around the neutrophil nadir at 7 to 14 days after a cycle, and defer elective care when counts are low. Transplant recipients defer elective care for three to six months, and plaque control reduces cyclosporine-induced gingival enlargement.

#494 · Clinical Therapy

What is the osteoradionecrosis risk after head and neck radiotherapy?

Answer

It is lifelong, higher in the mandible, and rises sharply above about 60 Gy. Extractions are done two to three weeks before radiation, and afterward any extraction in the irradiated field needs the radiation oncologist's input. These clients need daily 1.1 percent sodium fluoride, three-month recalls and xerostomia management.

#495 · Clinical Therapy

What risk do antiresorptive medications carry, and how do you respond?

Answer

Bisphosphonates, denosumab and antiangiogenic agents risk medication-related osteonecrosis of the jaw, exposed bone persisting more than eight weeks without prior jaw radiation. Risk is far lower at oral osteoporosis doses than intravenous oncology doses. Non-surgical hygiene care is safe and protective, and drug holidays are not routinely indicated.

#496 · Clinical Therapy

How do you manage a client with a latex allergy?

Answer

Separate immediate IgE-mediated reactions from delayed contact dermatitis, and ask about banana, avocado, kiwi and chestnut cross-reactivity. Book the first appointment of the day, and check gloves, rubber dam and prophy cups for latex before the client is seated.

#497 · Clinical Therapy

What do you do if a client has a seizure during treatment?

Answer

Stop treatment, clear the mouth, lower the client to a supported supine position, do not restrain them or place anything in the mouth, turn the head to the side and time the event. Beyond five minutes is status epilepticus needing emergency services. Record seizure type, triggers and any phenytoin-induced gingival enlargement.

#498 · Clinical Therapy

How does whitening work, and when do you use caution or decline?

Answer

Hydrogen or carbamide peroxide releases free radicals that oxidize chromogens. Transient sensitivity is the commonest adverse effect, then gingival irritation from tray overflow. It does not lighten composite or ceramic, and bonding waits about two weeks. Use caution with untreated caries, exposed roots, cracked teeth, tetracycline staining and pregnancy.

#499 · Clinical Therapy

How is silver diamine fluoride used, and what must consent cover?

Answer

At 38 percent it delivers about 44 800 ppm fluoride and arrests active caries when applied twice yearly. Consent must cover permanent black staining of the arrested lesion. Avoid it with silver allergy, ulcerative stomatitis or pulpal involvement. Protect soft tissue with petroleum jelly, dry the lesion, apply for one minute and do not rinse.

#500 · Clinical Therapy

How do you instrument and maintain implants and prostheses?

Answer

Peri-implant mucositis is reversible inflammation with bleeding on probing and no bone loss; peri-implantitis adds bone loss. Probe gently against a baseline, use titanium, plastic, resin or graphite instruments rather than stainless steel, and prefer glycine air-polishing powder and neutral fluoride, since acidulated fluoride etches titanium. Teach floss threaders under pontics and nightly denture removal.